

Galois Theory

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1 Extension of Fields

§1.1 Introduction

Definition 1.1.1. Consider a field K . Let $\text{Aut}(K)$ be the group of **automorphisms** of K , with its set as

$$\{g: K \rightarrow K \mid g(x+y) = g(x) + g(y), g(xy) = g(x)g(y), \forall x, y \in K\},$$

and the group operation being the composition of maps.

Definition 1.1.2. $g: K \rightarrow K$ is an automorphism if g has an inverse.

Example 1.1.3 ($\text{Aut}(\mathbb{Q})$)

Take any $g \in \text{Aut}(\mathbb{Q})$. Then, $g(1) = 1$. Next, we get that $g(2) = g(1) + g(1) = 1$. By induction, we get $g(n) = n$. Using this, we can get $g(0) = g(n + (-n)) = 0$. Therefore, g fixes all integers. Using this, we can conclude that g fixes all fractions. Concluding, we get that

$$\text{Aut}(\mathbb{Q}) = \{e\}.$$

Example 1.1.4 ($\text{Aut}(\mathbb{Q}(\sqrt{2}))$)

Recall that $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$. Now, let $g \in \text{Aut}(\mathbb{Q}(\sqrt{2}))$. Then,

$$x^2 - 2 = 0 \implies g(x^2 - 2) = 0 \implies g(x)^2 - 2 = 0$$

Then, either $g(\sqrt{2}) = \sqrt{2}$ or $g(\sqrt{2}) = -\sqrt{2}$. In the first case, $g = id$, and in the latter, $g \neq id$. Thus, we can conclude that

$$\text{Aut}(\mathbb{Q}(\sqrt{2})) \cong \mathbb{Z}/2.$$

In general, if $f(x)$ is a polynomial of degree n over a field, its *Galois group* is a subgroup of S_n , the group of permutations of $\{1, \dots, n\}$. It turns out that if an equation is solvable by radicals, then the Galois group of this polynomial is solvable. This means that G has a descending chain of subgroups $G \supset G_1 \supset \dots$ such that each G_i is a normal subgroup of G_{i-1} , and G_{i-1}/G_i is a cyclic group.

Example 1.1.5

S_3 is solvable:

$$S_3 \supset \{(123), (132), e\} \supset \{e\}$$

$\cong \mathbb{Z}/3$ and normal

Example 1.1.6

S_4 is solvable as well.

Example 1.1.7

S_5 is not solvable. Only A_5 is a non-trivial normal subgroup of S_5 . Further, A_5 is simple, so it doesn't have any normal subgroups, so we can't build a chain. The *general* quintic has Galois group S_5 , so it can't be solved by radicals.

§1.2 Extensions of Fields

To begin with, note that because any homomorphism of fields is injective, every homomorphism of fields is an embedding (smaller one in bigger one). Note that this is true because the kernel is an ideal, and all homomorphisms send 1 to 1, and since fields only have the zero ideal or itself, the kernel is equal to the zero ideal.

Definition 1.2.1. Define $\text{Emb}(K, L)$ to be the set of **embeddings** of $K \hookrightarrow L$ for embeddings of K into L .

Remark 1.2.2 — Note that $\hookrightarrow$ means that it is an injective map.

Example 1.2.3

$\text{Aut}(\mathbb{Q}) = \text{Emb}(\mathbb{Q}, \mathbb{Q}) = \{e\}$.

Now recall the definition of a characteristic; it is either equal to a prime number p or equal to 0.

Example 1.2.4

All finite fields have positive characteristic. However, also $\mathbb{F}_p(t)$ has positive characteristic.

Example 1.2.5

$\text{char}(\mathbb{Q}) = 0$. Note that any field with characteristic 0 has a subfield isomorphic to $\mathbb{Q}$.

Remark 1.2.6 — Note that $\mathbb{F}_p$ and $\mathbb{Q}$ are the smallest fields of positive and zero characteristic, respectively.

Let $K \subseteq L$ be fields; that is, we have an extension of fields. Then, L is a vector space over K .

Definition 1.2.7. An extension $K \subseteq L$ is called **finite** if L is a finite dimensional vector space over K . We call $\dim_K(L)$ the **degree** of the extension. We write $[L : K] = \dim_K(L)$.

Recall the following theorem.

Theorem 1.2.8 (Tower Law)

Suppose we have $F \subseteq K \subseteq L$ are finite extension of fields. Then,

$$[L : F] = [L : K][K : F].$$

Remark 1.2.9 — Not all extensions are finite. For instance, $k \subseteq k(t)$, where t is a free variable, is not a finite extension.

Exercise 1.2.10. Let $K \subset L$ be a finite field extension. Then, show that if R is a ring such that $K \subset R \subset L$, then R is a field.

Solution: Let $a \in R$ be nonzero. Since R is finite dimensional, we have a minimal polynomial $a_0 + \cdots + x^n$ for a . This is irreducible, so $a_0 \neq 0$. Then, if we shift over the a_0 and divide by a_0 , we get an inverse for a :

$$a_0 = -a_1a - \cdots - a_{n-1}a^{n-1} - a^n = a(-a_1 - \cdots - a_{n-1}a^{n-1}),$$

so

$$1 = a \left(a_0^{-1}(-a_1 - \cdots - a_{n-1}a^{n-1}) \right).$$

□

Exercise 1.2.11. Let F_1, F_2 be fields between K and L . Show that there is a smallest subfield F_1F_2 (called the **compositum**) of L containing both F_1 and F_2 . Show that this is spanned by elements of the form $\sum_{i=1}^n \lambda_i \mu_i$, for $\lambda_i \in F_1$, $\mu_i \in F_2$.

Solution: It is clear that $F_1F_2 = K(F_1, F_2)$ (abuse of notation). Then, we know that there are elements $a_1, \dots, a_m \in F_1$ and $b_1, \dots, b_n \in F_2$ such that

$$F_1F_2 = K[a_1, \dots, a_m, b_1, \dots, b_n].$$

It is then clear that every polynomial expression in the a_i and b_j can be written as a finite sum. □

§1.3 Polynomials

Let $K \subseteq L$ be a finite extension of fields. Take any $\alpha \in L$. Consider $1, \alpha, \alpha^2, \dots$. Suppose that n is the smallest natural number such that $1, \alpha, \dots, \alpha^{n-1}$ are linearly dependent over K . First note that if $d = [L : K]$, then every $d + 1$ elements of L are linearly dependent over K . Thus, such an n exists. Then, $\exists a_0, a_1, \dots, a_{n-1} \in K$ not all equal to 0 such that

$$a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0, \quad a_{n-1} \neq 0.$$

WLOG, we can assume that $a_{n-1} = 1$. Hence, we've found a polynomial $f(x) \in K[x]$, monic of least degree such that $f(\alpha) = 0$. This is called the **minimal polynomial** of α .

Note that in a field K , we can consider $K[x]$. This is a Euclidean domain, so $K[x]$ is a PID. In fields, we have:

prime ideals = maximal ideals = ideals generated by irreducible polynomials

Therefore, if $f(x) \in K[x]$ is irreducible, then $K[x]/(f(x)) =: L$ is a field. Firstly, we get that L has a basis consisting of the images of $1, x, \dots, x^{d-1}$, where $d = \deg(f(x))$. Writing

$$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_0,$$

this becomes zero in the quotient, so

$$x^d = -(a_{d-1}x^{d-1} + \cdots + a_0)$$

in L . From this, we get a few crucial properties:

- $[L : K] = d$.
- Let $\alpha \in L$ be the image of x . Then, $f(\alpha) = 0$.

For the second, indeed $f(x)$ goes to 0 in L , so $f(\alpha) = 0$.

Example 1.3.1

$d = 2$, and let $f(x) = x^2 - a$, for $a \in K$. $K[x]/(x^2 - a) = K \cdot 1 + K \cdot x$. a is not a square in K , then $[L : K] = 2$.

Definition 1.3.2. Let $K \subseteq L$ be a finite extension of fields, $\alpha \in L$. We define $K(\alpha)$ to be the smallest subfield of L that contains K and α . This is just the intersection of all subfields of L that contain K and α .

Proposition 1.3.3

$K(\alpha) \cong K[x]/(f(x))$, where $f(x)$ is the minimal polynomial of α .

Proof. Consider $\phi: K[x] \rightarrow K(\alpha)$ sending x to α is surjective. We first claim that ϕ is surjective. Note that $K(\alpha)$ contains $1, \alpha, \dots, \alpha^{d-1}$, where $d = \deg(f(x))$ and that $f(\alpha) = 0$. $K(\alpha) = K \cdot 1 + K \cdot \alpha + \dots + K \cdot \alpha^{d-1}$ is already a field. We want to show that it is closed under division now. Any nonzero element can be written as $b_{d-1}\alpha^{d-1} + \dots + b_1\alpha + b_0$. Thus, $\exists$ polynomial $g(x)$ of $\deg(g(x)) < d$ such that my element is $g(\alpha)$. Then, the polynomials $f(x)$ and $g(x)$ are coprime (due to irreducibility of f and the degree), hence there exists polynomials $r(x), s(x)$ in $K[x]$ such that

$$1 = r(x)f(x) + s(x)g(x). \\ = 0 \text{ in } L$$

Therefore, in L , $1 = s(\alpha)g(\alpha)$, implying $g(\alpha)^{-1} = s(\alpha)$. Thus, this field is closed in taking multiplicative inverses.

Next, we want to find the kernel of ϕ . This is the set of all polynomials vanishing at α . This set consists of all multiples of $f(x)$, $(f(x))$. Thus, injective.

Therefore, we get

$$K[x]/(f(x)) \xrightarrow{\sim} K(\alpha).$$

□

Definition 1.3.4. Let $K \subseteq L$ be an extension of fields, and suppose $\alpha_1, \dots, \alpha_n \in L$. Define $K(\alpha_1, \dots, \alpha_n)$ to be the smallest subfield containing of L contains K and $\alpha_1, \dots, \alpha_n$.

Using the above, $K(\alpha_1) \subseteq K(\alpha_1, \dots, \alpha_n)$.

$$K(\alpha_1) \cdot 1 + K(\alpha_1) \cdot \alpha_2 + \dots = K(\alpha_1, \alpha_2).$$

Iterating, we see that $K(\alpha_1, \dots, \alpha_n)$ is a vector space over K spanned by finitely many monomials $\alpha_1^{\Gamma_1} \dots \alpha_n^{\Gamma_n}$,

Lemma 1.3.5

Let $k \subseteq K$ be a finite field extension. Then, there exists $\alpha_1, \dots, \alpha_n \in K$ such that $K = k(\alpha_1, \dots, \alpha_n)$. In this case, we can say that $\alpha_1, \dots, \alpha_n$ **generate** K .

Proof. Assume $k \neq K$. Then, there exists an element $\alpha_1 \in K \setminus k$. Consider $k(\alpha_1) \subset K$. If $k(\alpha_1) = K$, then stop. Otherwise, $\exists \alpha_2 \in K \setminus k(\alpha_1)$. Form $k(\alpha_1)(\alpha_2) = k(\alpha_1, \alpha_2)$, and so the process will terminate after finitely many steps. Indeed, $[K : k] = [K : k(\alpha_1)][k(\alpha_1) : k]$. Since $\alpha_1 \notin k$, we have $[k(\alpha_1) : k] \geq 2$. This is at least 2 because the degree of the minimal polynomial of α_1 over k is at least 2. Thus, $[K : k(\alpha_1)] < [K : k]$. By induction, $K = k(\alpha_1)(\alpha_2, \dots, \alpha_n) = k(\alpha_1, \dots, \alpha_n)$. $\square$

Remark 1.3.6 — All homomorphisms of fields are injective; that is, these are embeddings.

Example 1.3.7

$\text{Aut}(\mathbb{Q}(\sqrt{a})) = \{id, \text{'conjugation'}\}$, where a is not a square in $\mathbb{Q}$.

Suppose $a \in \mathbb{Q}$, but a is not a square in $\mathbb{Q}$. Define $\phi: \mathbb{Q}(\sqrt{a}) \rightarrow \mathbb{Q}(\sqrt{a})$ such that $\phi(x) = x$, for all $x \in \mathbb{Q}$. Now, we consider where $\phi(x + y\sqrt{a})$ goes to.

$$\phi(x + y\sqrt{a}) = \phi(x) + \phi(y\sqrt{a}) = x + \phi(y)\phi(\sqrt{a}) = x + y\phi(\sqrt{a}).$$

Thus, we want to know where ϕ send $\sqrt{a}$. $\sqrt{a}$ is a root of $t^2 - a = 0$.

$$\phi(\sqrt{a})^2 - a = 0 \implies \phi(\sqrt{a}) = \pm\sqrt{a}.$$

If $\phi(\sqrt{a}) = \sqrt{a}$, then $\phi = id$. If $\phi(\sqrt{a}) = -\sqrt{a}$, then ϕ is the ‘conjugation’.

Example 1.3.8

$\text{Emb}(\mathbb{Q}, \mathbb{Q}(\sqrt{a}))$ has only one element: the tautological map $\phi(x) = x$, $x \in \mathbb{Q}$.

Note that the set of embeddings may be empty as well:

Example 1.3.9

$\text{Emb}(\mathbb{Q}(\sqrt{a}), \mathbb{Q}(\sqrt{b}))$ is empty if $a \neq bc^2$, for some $c \in \mathbb{Q}$.

We know that $\phi(\sqrt{a})$ goes to a root of $t^2 - a = 0$ in the field $\mathbb{Q}(\sqrt{b})$. We check that $t^2 - a = 0$ has no roots in $\mathbb{Q}(\sqrt{b})$. Indeed,

$$(x + y\sqrt{b})^2 = a \iff \begin{cases} x^2 + by^2 = a \\ 2xy = 0 \end{cases}.$$

Thus $x = 0$ or $y = 0$. if $y = 0$, then the first equation can't be solved since a is not a square in $\mathbb{Q}$. If $x = 0$, then this cannot occur due to our assumption, so we get a contradiction in both cases.

Definition 1.3.10. Suppose that $k \subseteq L$ is a field extension, and $k \subseteq M$ is another field extension. Define $\text{Emb}_k(L, M)$ to be the set of all embeddings of L into M which are the identity map on k .

Example 1.3.11

Let $L = \mathbb{R}$, $M = \mathbb{C}$, $k = \mathbb{Q}$. Then, $\text{Emb}_{\mathbb{Q}}(\mathbb{C}, \mathbb{C})$ (which is just $\text{Aut}_{\mathbb{Q}}(\mathbb{C})$ or just $\text{Aut}(\mathbb{C})$ since all rationals fix everything in $\mathbb{C}$) is huge.

However, $\text{Emb}_{\mathbb{R}}(\mathbb{C}, \mathbb{C}) = \text{Aut}_{\mathbb{R}}(\mathbb{C}) = \{id, c\}$, where c is complex conjugation (by the same argument). Note that $\mathbb{C} = \mathbb{R} \oplus \mathbb{R}i$, $[\mathbb{C} : \mathbb{R}] = 2$.

Remark 1.3.12 — Suppose $K \subseteq L$ and $K \subseteq M$ are field extensions, and $[L : K] < \infty$. Then $\text{Emb}_K(L, M)$ is a finite set.

Proof. By lemma 1.3.5, we can write $L = K(\alpha_1, \dots, \alpha_n)$, for some $\alpha_i \in L$. Suppose $\xi: L \rightarrow M$ is a **K -linear embedding**. That is, the restriction of ξ to K is the identity map $K \rightarrow K$. Let $f_i(x)$ be the minimal polynomial of α_i over K . We want to find the possibilities for ξ . We have that $f_i(\alpha_i) = 0$ in L . Applying ξ , and writing $f_i(\alpha_i) = \alpha_i^d + a_{i-1}\alpha_i^{d-1} + \dots + a_0$, we get

$$0 = \xi(f_i(\alpha_i)) = f_i(\alpha_i) = \xi(\alpha_i)^d + a_{i-1}\xi(\alpha_i)^{d-1} + \dots + a_0 = f_i(\xi(\alpha_i)).$$

Thus, ξ must map a root of f to another root of f . Therefore, there are only finitely many possibilities for $\xi(\alpha_i)$, for each i , proving the result. $\square$

Definition 1.3.13. Define $\text{Aut}_K(L) = \text{Emb}_K(L, L)$, where $K \subseteq L$. This is the **automorphism group** of L over K .

Remark 1.3.14 — Note that if $[K : L]$ is finite, then we have that any K -linear embedding from L to L is actually an automorphism. This follows by the rank nullity theorem, if we view L as a finite dimensional K -vector space. This is not the case for an infinite extension L/K .

Proposition 1.3.15

Let k be a field. Suppose $k \subseteq K$ and $k \subseteq L$ are field extensions such that K/k is a finite extension. Then, there is a field extension $k \subseteq M$ such that both K and L have k -linear embeddings into M . Moreover, we can ensure that M can be chosen to be finite over L .

Proof. We induct on $d = [K : k]$. If $d = 1$, i.e., $K = k$, we just take $M = L$. Now assume $d \geq 2$. Then, we can find some $\alpha \in K \setminus k$. Let $f(x)$ be the minimal polynomial of α over k . Consider $k' = k[x]/(f(x))$. This field is isomorphic to $k(\alpha)$; the map sends x to α . Note that $[k' : k] = \deg(f(x)) \geq 2$, but the tower law implies that $[K : k'] < d$.

Consider $f(x)$ as an element of $L[x]$. Let $g(x) \in L[x]$ be an irreducible factor of $f(x)$ in $L[x]$. Consider $L' = L[x]/(g(x))$. This is a field extension of L . Denote by b the image

of x in L' . Thus, $g(b) = 0$, so $f(b) = 0$. We have a natural map $k[x] \hookrightarrow L[x] \twoheadrightarrow L'$ such that $h(x) \mapsto h(b)$ (the evaluation at b), where $h(x) \in k[x]$ and $h(b) \in L'$. The kernel consists of all polynomial in $k[x]$ that vanish at b , e.g. $f(b) = 0$. Thus, $f(x)$ is in the kernel. However, $k[x]$ is a PID, so the kernel is generated by a some monic polynomial, say $h(x)$. Thus, $h(x) \mid f(x)$. Since $f(x)$ is irreducible and monic (by definition), we have $h(x) = f(x)$. Thus, the kernel is generated by $f(x)$. Recall that if $R_1 \xrightarrow{\phi} R_2$ is a homomorphism of rings, then we have an injective homomorphism $R_1 / \ker(\phi) \hookrightarrow R_2$. We get an injective homomorphism $k' \hookrightarrow L'$. Note that $k[x] \twoheadrightarrow k'$ and $k' \cong k(\alpha) \hookrightarrow K$, which has a map of degree less than d . By the inductive assumption, we can find $k' \hookrightarrow M$, some field, such that k' and L' embed k' -linearly into M . This M does the job, because a k' -linear embedding is also k -linear. $\square$

Remark 1.3.16 — In the above, we note that $\twoheadrightarrow$ means surjective.

Exercise 1.3.17. Let $k \subset K$ be a field extension. Let $f(x)$ and $g(x)$ be polynomials in $k[x]$ and let $h(x)$ be the gcd of $f(x)$ and $g(x)$ in $k[x]$. Prove that $h(x)$ is the gcd of $f(x)$ and $g(x)$ in $K[x]$.

Remark 1.3.18 — What this exercise is saying is that the gcd is unique regardless of what field we are working in.

2 Normal Extensions

Definition 2.0.1. Let k be a field and let $f(x) \in k[x]$ be monic. A finite field extension $k \subset K$ is called a **splitting field** of $f(x)$ if the following hold:

- $f(x)$ is **completely split** in K , i.e. $f(x) = \prod_{i=1}^n (x - \alpha_i)$, for some $\alpha_i \in K$.
- $K = k(\alpha_1, \dots, \alpha_n)$.

Remark 2.0.2 — We (implicitly) don't consider constant polynomials for splitting fields.

Remark 2.0.3 — It is implicit in the second condition that this extension is minimal.

Proposition 2.0.4

Every polynomial has a splitting field. All splitting fields of $f(x) \in k[x]$ are k -linearly isomorphic.

Proof. We induct on $d = \deg(f(x))$. If $d = 1$, the root of $f(x)$ is in k , so k is a splitting field. Now suppose $d \geq 2$. Choose an irreducible factor $g(x)$ of $f(x)$. Define $L = k[x]/(g(x))$ is a field extension of k . Let α be the image of x in L . Then, $g(\alpha) = 0$, so $f(\alpha) = 0$. We have $\frac{f(x)}{x-\alpha} \in L[x]$. $\deg\left(\frac{f(x)}{x-\alpha}\right) = d - 1$, so we can apply the induction assumption and conclude that there is a finite field extension $L \subset K$ such that $\frac{f(x)}{x-\alpha}$ completely splits over K and K is generated by the roots of $\frac{f(x)}{x-\alpha}$. Then, K is a splitting field of $f(x)$. K is generated by the roots of $f(x)$ over L , i.e. all elements of K are sums of $\alpha_1^{r_1} \dots \alpha_n^{r_n}$ with coefficients in L (here α_i s are the roots of $f(x)$). However, $L = k(\alpha)$ ($\alpha = \alpha_1$), so L is also generated by the roots of $f(x)$ over k . Therefore, K is generated by the roots of $f(x)$ over k .

Now for uniqueness (isomorphism part). Suppose $k \subset K$ and $k \subset L$ are splitting fields of $f(x)$. By proposition 1.3.15, there exists an extension $k \subset M$ such that both K and L embed k -linearly into M . That is, if we have $k \xrightarrow{i} K \xrightarrow{\phi} M$ and $k \xrightarrow{j} L \xrightarrow{\psi} M$, we have $\phi i = \psi j$. Since ϕ is injective, $\phi(K) \cong K$. Since ψ is injective, $\psi(L) \cong L$. In K , we have

$$f(x) = \prod_{i=1}^n (x - \alpha_i),$$

where $\alpha_i \in K$. ϕ is a k -linear embedding, i.e. ϕ is the identity on k . Thus, $\phi(f(x)) = f(x)$, because $\phi(a_i) = a_i$ where $f(x) = \sum_{i=0}^n a_i x^i$. Since $f(x)$ completely splits in K , $f(x)$ completely splits in M . $f(x) = \prod_{i=1}^n (x - \phi(\alpha_i)) \in M[x]$. We know that $K = k(\alpha_1, \dots, \alpha_n)$. Therefore, $\phi(K) = k(\phi(\alpha_1), \dots, \phi(\alpha_n))$. The same applies to L and ψ . Let $\beta_1, \dots, \beta_n$ be

the roots of $f(x)$ in M , so we have $f(x) = \prod_{i=1}^n (x - \beta_i)$ over M . Consider $k(\beta_1, \dots, \beta_n) \subset M$. Since $\{\beta_1, \dots, \beta_n\} = \{\phi(\alpha_1), \dots, \alpha(n)\}$, we have $k(\beta_1, \dots, \beta_n) = \phi(K)$. For the same reason, $\psi(L) = k(\beta_1, \dots, \beta_n)$. Thus, $\phi(K) = \psi(L)$. Now, we have isomorphisms $K \cong \phi(K) = \psi(L) \cong L$. $\square$

Definition 2.0.5. An extension of fields $k \subset L$ is **normal** if for any k -linear embeddings $\psi_1, \psi_2: L \hookrightarrow M$, there is a k -automorphism $\sigma: L \rightarrow L$ (which is a k -linear embedding of L into L) such that $\psi_2 = \psi_1 \circ \sigma$.

Example 2.0.6

Take $k = \mathbb{Q}$ and $L = \mathbb{Q}[x]/(x^2 - 2)$. This extension is normal.

Take $M = \mathbb{C}$ (or $\mathbb{R}$). Then, ψ_1 is induced by $\mathbb{Q}[x] \rightarrow \mathbb{C}$ given by evaluating at $\sqrt{2} > 0$. What we mean by induced is that there are the maps $\mathbb{Q}[x] \rightarrow L \xrightarrow{\psi_1} \mathbb{C}$. We also let ψ_2 to be induced by $\mathbb{Q}[x] \rightarrow \mathbb{C}$ given by evaluating at $-\sqrt{2} < 0$. We claim that there exists $\sigma \in \text{Aut}_k(L)$ such that $\psi_2 = \psi_1 \sigma$. Indeed, $\sigma(x) = -x$. σ is the identity on $\mathbb{Q}$.

Remark 2.0.7 — We can run a similar argument to show that any quadratic extension over a field of characteristic not 2 is normal.

Example 2.0.8

Take $k = \mathbb{Q}$ and $L = \mathbb{Q}[x]/(x^3 - 2)$. Define ψ_1 be induced by evaluating x at $\sqrt[3]{2}$, the unique real cube root of 2. Let ψ_2 be induced by evaluating x at $\sqrt[3]{2} \frac{-1 + \sqrt{3}i}{2}$. We claim that there is no automorphism $\sigma \in \text{Aut}_{\mathbb{Q}}(L)$ such that $\psi_2 = \psi_1 \sigma$.

We have $\psi_1: L \xrightarrow{\sim} \mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R} \subset \mathbb{C}$ and $\psi_2: L \xrightarrow{\sim} \mathbb{Q}(\sqrt[3]{2} \frac{-1 + \sqrt{3}i}{2}) \subset \mathbb{C}$. If $\psi_2 = \psi_1 \sigma$, then $\psi_2(L) = \psi_1(L)$. However, our two maps are not equal in $\mathbb{C}$, so we get a contradiction. Therefore, we can conclude that our extension is not normal.

Remark 2.0.9 — We can run a similar argument to show that any extension of higher degree is not normal.

Remark 2.0.10 — $k \subset L$ is normal if and only if $\psi_1(L) = \psi_2(L)$ for any k -linear embeddings $\psi_1, \psi_2: L \hookrightarrow M$.

Proof. ($\implies$): We have $\psi_2 = \psi_1 \circ \sigma$, so $\psi_1(L) = \psi_2(L)$.

($\impliedby$): We have two embeddings $L \xrightarrow[\sim]{\psi_1} \psi_1(L) \subset M$ and $L \xrightarrow[\sim]{\psi_2} \psi_2(L) \subset M$, where $\psi_1(L) = \psi_2(L)$. We now want a σ going from $L \rightarrow L$, but this is apparent: σ is defined as the unique map making the square commute. It's automatically a k -linear isomorphism of L with L . $\square$

Remark 2.0.11 — $k \subset L$ is normal if and only if the map natural injective map compose with some fixed embedding $L \hookrightarrow M$: $\text{Aut}_k(L) \rightarrow \text{Emb}_k(L, M)$ is bijective.

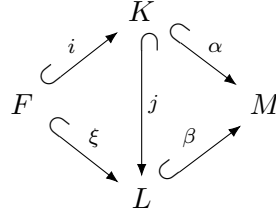
Proof. ($\implies$): Fix $\tau \in \text{Emb}_k(L, M)$. Then, since L/k is normal, we get that $\tau = \tau' \circ \sigma$ for some $\sigma \in \text{Aut}_k(L)$, so it's surjective. For injectivity, it's a composition of an injective map and an isomorphism, so it's injective.

($\impliedby$): If this map is bijective, then it's necessarily surjective, so we get that for $\psi_1, \psi_2: L \hookrightarrow M$, there's an automorphism $\sigma \in \text{Aut}_k(L)$ such that $\psi_1 = \psi_2 \circ \sigma$. $\square$

Lemma 2.0.12 ('No-Escape' Lemma)

Let $k \hookrightarrow F \xrightarrow{i} K \xrightarrow{j} L$ be finite field extensions, where i and j are k -linear. Suppose that K is normal over k . Then, any k -linear embedding $\xi: F \hookrightarrow L$ is equal to the composition $\xi = j\sigma i$, where $\sigma \in \text{Aut}_k(K)$. In particular, the image of any $F \hookrightarrow L$ is contained in K . Thus, F cannot escape from K .

Proof. By proposition 1.3.15, we can simultaneously extend $i: F \hookrightarrow K$ and $\xi: F \hookrightarrow L$ to an embedding $F \hookrightarrow M$, that is, there are embeddings $\alpha: K \hookrightarrow M$ and $\beta: L \hookrightarrow M$ such that $\alpha i = \beta \xi$. That is, we have a commutative diagram. Since K is normal, by definition, $\exists \sigma \in \text{Aut}_k(K)$ such that $\alpha = \beta j \sigma$. This implies that $\alpha i = \beta j \sigma i$, but by the commutativity of the square, we have $\beta \xi = \alpha i$. Now, all these maps are injective, and so $\beta(x) = \beta(y) \implies x = y$. Therefore, $\beta \xi = \beta j \sigma i$ implies $\xi = j \sigma i$.



$\square$

Remark 2.0.13 — The point of the no-escape lemma is that any embedding from F to L needs to go through K (in a very vague sense).

Corollary 2.0.14

Let $k \subset F \xrightarrow{i} K \xrightarrow{j} L$ be finite field extensions, where K/k is normal. Then, the map $\text{Aut}_k(K) \rightarrow \text{Emb}_k(F, L)$ given by 'compose with j and precompose with i '

$$F \xrightarrow{i} K \xrightarrow{\sigma} K \xrightarrow{j} L$$

is surjective.

Proof. This is an immediate consequence of lemma 2.0.12. $\forall \xi \in \text{Emb}_k(F, L)$, we write it as $\xi = j\sigma i$. $\square$

Remark 2.0.15 — Here is an often scenario where we apply this. Take $K = L$. Then, corollary 2.0.14 gives that any k -linear embedding F into K (normal over k) is obtained from one such embedding by composing with a k -linear automorphism of K .

$$k \subset F \hookrightarrow K.$$

In other words, all k -linear embeddings of F into a normal extension K differ by k -linear automorphisms of K .

Theorem 2.0.16 (Main Theorem of Normal Extensions)

Let $k \subset K$ be a finite extension of fields. Then, the following are equivalent:

1. K/k is normal.
2. K is a splitting field of some polynomial $f(x) \in k[x]$.
3. For any irreducible polynomial $g(x) \in k[x]$ either $g(x)$ has no roots in K or $g(x)$ completely splits in K .

Proof. (2 $\implies$ 1): Assume that K is a splitting field of $f(x)$. Let L be an extension of k and let $\psi: K \hookrightarrow L$ be a k -linear embedding. It's enough to show that $\psi(K)$ doesn't depend on the choice of ψ . Note $f(x)$ completely splits in K , hence also in $\psi(K)$, and also in L . Take the roots of $f(x)$ in L , which we'll call $\alpha_1, \dots, \alpha_n$. Consider $k(\alpha_1, \dots, \alpha_n)$. This is a splitting field for $f(x)$, so $\psi(K) = k(\alpha_1, \dots, \alpha_n)$. Therefore, $\psi(K)$ doesn't depend on ψ .

(3 $\implies$ 2): We must show that K is a splitting field of some $f(x) \in k[x]$. Since K/k is finite, we can find $\alpha_1, \dots, \alpha_n \in K$ such that $K = k(\alpha_1, \dots, \alpha_n)$. Let $f_i(x) \in k[x]$ be the minimal polynomial of α_i , for each i . By the assumption of 3., $f_i(x)$ splits completely in K because $f_i(\alpha_i) = 0$. Now take $f(x) = \prod_{i=1}^n f_i(x)$. This works.

(1 $\implies$ 3): We now assume K/k is normal. Let $g(x) \in k[x]$ be a polynomial such that $g(\alpha) = 0$ for some $\alpha \in K$. Let L/k be a splitting field of $g(x)$. Proposition 1.3.15 ensures that there is an extension $k \subset M$ such that both K and L embed k -linearly into M . Let β be any root of $g(x)$ in M . We need to show that $\beta \in K$. Consider the field $k[x]/(g(x))$. We know that $k(\beta) \cong k[x]/(g(x)) \cong k(\alpha)$. Let $F = k[x]/(g(x))$. Thus, $F \hookrightarrow K$ by sending the image of x in F to α (indeed, $k(\alpha) \subset K$). We also have $F \hookrightarrow M$ whereby the image of x goes to β . By lemma 2.0.12, F cannot escape from K (since K is normal), i.e. the image of any embedding $F \hookrightarrow M$ is contained in K . Therefore, $\beta \in K$. $\square$

Remark 2.0.17 — The above theorem tells us that an extension is normal if and only if it contains all the roots of a polynomial.

Remark 2.0.18 — Note that if K/k is normal and L/K is normal, this does not imply L/k to be normal. The example below highlights this.

Example 2.0.19

Consider $L = \mathbb{Q}(\sqrt[4]{2})$, $K = \mathbb{Q}(\sqrt{2})$, $k = \mathbb{Q}$. Then, K/k is normal since it's a degree two extension. Further, L/K is normal because it's also a degree two extension. However, L/k is not a normal extension. We see that $L \cong \mathbb{Q}[x]/(x^4 - 2)$, but x^4 does not have all of its roots contained in L ; thus, L is not a splitting field for $x^4 - 2$.

Corollary 2.0.20

Let $k \subset L \subset K$ be finite extensions of fields. If K is normal over k , then K is also normal over L .

Proof. By the main theorem of normal extensions (2.0.16), there is a polynomial $f(x) \in k[x]$ such that K is a splitting field. We claim that K is a splitting field of $f(x)$ over L . Of course, $f(x)$ is completely split in K . Likewise, $K = k(\alpha_1, \dots, \alpha_n)$, where $f(x) = \prod_i (x - \alpha_i)$, so this implies that $K = L(\alpha_1, \dots, \alpha_n)$. Therefore our claim is proven. We can then use the main theorem to attain that K/L is normal. $\square$

Proposition 2.0.21

If $K \subset L$, $K \subset M$ are normal, then $K \subset L \cap M$ is normal.

Proof. Let $f(x) \in K[x]$ be irreducible and have a root in $L \cap M$. Then, $f(x)$ has a root in both L and M , so splits completely in both L and M . Therefore, all the roots of $f(x)$ in some bigger extension of L and M , say N , are in $L \cap M$, and so $L \cap M$ is normal. $\square$

Example 2.0.22

Consider $\mathbb{Q} \subset \mathbb{Q}(\sqrt[3]{2})$, where $\sqrt[3]{2} \in \mathbb{R}$. We've seen that this is not a normal extension. But, we can embed $\mathbb{Q}(\sqrt[3]{2})$ into $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$. The point is that $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$ is normal over $\mathbb{Q}$. By the main theorem for normal extensions, it's enough to show that $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$ is a splitting field of $x^3 - 2$, but this is apparent (by picture).

Proposition 2.0.23

Let $k \subset L$ be a finite field extensions. There exists a polynomial $f(x) \in k[x]$ such that L embeds k -linearly into a splitting field of $f(x)$. In particular, any finite extensions can be embedded k -linearly into a normal extension.

Proof. Take $\alpha_1, \dots, \alpha_n \in L$ such that $L = k(\alpha_1, \dots, \alpha_n)$. Let $f_i(x)$ be the minimal polynomial of α_i , for each i . Take $f(x) = \prod_{i=1}^n f_i(x)$. This will do. Let M be a splitting field of $f(x)$ over L . Let's show that M is a splitting field of $f(x)$ over k . $f(x)$ is completely split in M . Further, M is generated by the roots of $f(x)$ over L . But, since $L = k(\alpha_1, \dots, \alpha_n)$ and $\alpha_1, \dots, \alpha_n$ are roots of $f(x)$, so M is generated by the roots of $f(x)$ over k . $\square$

If L/K is a field extension, we associate to it the group $\text{Aut}_K(L)$. Then, normal sub-extensions correspond to normal subgroups.

Proposition 2.0.24

Let $K \subset L \subset M$ be a finite extension of fields. Suppose that L is normal over K . Then, $\text{Aut}_L(M)$ is a normal subgroup of $\text{Aut}_K(M)$. We have an injective homomorphism $\text{Aut}_K(M)/\text{Aut}_L(M) \rightarrow \text{Aut}_K(L)$. Moreover, if M is normal over K , then this is an isomorphism.

Proof. We denote $L \xhookrightarrow{i} M$. Take any $\sigma \in \text{Aut}_K(M)$. Since L is normal over K , there exists a unique $\tau \in \text{Aut}_K(L)$ making the diagram below commutative, i.e. $\sigma i = i\tau$.

$$\begin{array}{ccc} L & \xhookrightarrow{i} & M \\ \downarrow \tau & & \downarrow \sigma \\ L & \xhookrightarrow{i} & M \end{array}$$

Then, sending $\sigma \mapsto \tau$ gives a map $\text{Aut}_K(M) \rightarrow \text{Aut}_K(L)$. The kernel of $\text{Aut}_K(M) \rightarrow \text{Aut}_K(L)$ consists of all automorphisms of M fixing L point-wise, which is precisely $\text{Aut}_L(M)$. Therefore, $\text{Aut}_L(M)$ is normal in $\text{Aut}_K(M)$, since every kernel is normal. Then, the injective map $\text{Aut}_K(M)/\text{Aut}_L(M) \rightarrow \text{Aut}_K(L)$ comes from usual group theory.

Now assume that M/K is also normal. Take any $\tau \in \text{Aut}_K(L)$. We have K -linear embeddings $i: L \hookrightarrow M$ and $i\tau: L \hookrightarrow M$. By corollary 2.0.14, there exists a $\sigma \in \text{Aut}_K(M)$ such that $i\tau = \sigma i$. Thus, we get a commutative diagram again (but for a different reason). Hence, the map $\text{Aut}_K(M)/\text{Aut}_L(M) \rightarrow \text{Aut}_K(L)$ is surjective if M/K is normal (since every τ comes from some σ). Therefore, we get this is an isomorphism. $\square$

3 Separable Extensions

§3.1 Irreducibility and Separability

Definition 3.1.1. A polynomial $f(x) \in k[x]$, where k is a field, is called **separable** if, in a splitting field $K \subset L$,

$$f(x) = \prod (x - \lambda_j) \in L[x],$$

where the roots λ_j are pairwise distinct (i.e., no roots with multiplicity).

Remark 3.1.2 — Note that this property is independent of the choice of splitting field.

Proposition 3.1.3 (Jacobian Criterion for Smoothness)

$f(x) \in k[x]$ is separable if and only if the $\text{hcf}(f, f') = 1$.

Proof. We work in $L[x]$, where L is a splitting field. $f(x)$ is not separable if and only if $(x - a)^2$ is a factor of $f(x)$, implying that $(x - a) \mid f'(x)$. This is equivalent to $(x - a) \mid \text{hcf}(f, f')$, so the highest common factor is not equal to 1. For the converse, we can show that $(x - a)^2 \nmid f(x)$. $\square$

Corollary 3.1.4

Let $f(x) \in k[x]$ be irreducible. Then:

1. If $\text{char}(k) = 0$, $f(x)$ is separable.
2. If $\text{char}(k) = p$, then f is inseparable if and only if $\exists g(x) \in k[x]$ that is irreducible such that $f(x) = g(x^p)$.

Proof. 1. In characteristic 0, we have $\deg(f') = \deg(f) - 1$, for $\deg(f) > 0$. Then, $\text{hcf}(f, f') = c$ must divide f, f' . Then, $c = 1$, since $c \neq f$.

2. Let $\text{hcf}(f, f') = c$ again. Then, $c \mid f$ means either $c = 1$, so it's separable, or $c = f$ and $c \mid f'$, implying $f' = 0$. Now, f' equals 0 if and only if $\exists g(x) \in k[x]$ such that $f(x) = g(x^p)$. $\square$

Example 3.1.5

Let $k = \mathbb{F}_p(T)$, the fraction field of $\mathbb{F}_p[T]$. Consider $f(x) = X^p - T \in k[X]$. We first note that $f(x)$ is irreducible by Eisenstein. Then, $f'(x) \equiv 0$, so $\text{hcf}(f, f') = f$. Thus, $f(x)$ is inseparable. Since $f(x)$ is irreducible, we can just make a root u of f and take the quotient $L = k[x]/(X^p - T)$. Then,

$$x = [x] = x + (X^p - T)$$

and $L = k(u)$ and $u^p = T$ in L . Note that $X^p - T = (X - u)^p$ in L . In fact, $X^p - T$ is *totally inseparable*, i.e., it only has one root.

Example 3.1.6

Let $k = \mathbb{F}_p(T)$, the fraction field of $\mathbb{F}_p[T]$. Consider $f(X) = X^p - TX - T \in k[X]$. This is separable.

§3.2 Starting the Theory

Definition 3.2.1. An extension $K \subset L$ is **separable**, if for every $a \in L$, the minimal polynomial $f(x) \in K[x]$ of a is separable.

We will now start this theory of separable extensions.

Definition 3.2.2. Let $K \subset L$ be a finite field extension. We define the **separable degree**, $[L : K]_S$, as the cardinality of $\text{Emb}_K(L, M)$, where $K \subset M$ is a normal field extension such that $\text{Emb}_K(L, M) \neq \emptyset$.

Remark 3.2.3 — The definition above is saying that there must be at least one embedding that goes from L to M .

Remark 3.2.4 — $[L : K]_S$ doesn't depend on the choice of M . Suppose that there are two extensions M_1, M_2 . Then, we can take some bigger extension M . As M_1 and M_2 are normal, the maps $\text{Emb}_K(L, M_1) \rightarrow \text{Emb}_K(L, M) \leftarrow \text{Emb}_K(L, M_2)$ are bijective by lemma 2.0.12 (L cannot escape from M_1 and M_2).

We first consider the case of primitive extensions. Suppose $f(x) \in K[x]$ is an irreducible polynomial. Then, we can form the primitive extension $K \subset K(a)$, where $f(a) = 0$. Now, suppose M is any field extension of K . We then have $\text{Emb}_K(K(a), M) = \{b \in M \mid f(b) = 0\}$, as $K(a) \cong K[x]/(f(x))$. Then, in this case, we know that $[K(a) : K]_S$ is equal to the number of roots in a splitting field.

Remark 3.2.5 — This discussion above tells us that $[K(a) : K]_S = \deg(f(x))$ if and only if f is separable.

Theorem 3.2.6 (Tower Law)

Let $K \subset L \subset M$ be finite field extensions. Then, we have

$$[M : K]_S = [M : L]_S [L : K]_S.$$

Proof. Let us denote the embedding $L \subset M$ by $\alpha: L \hookrightarrow M$. We can find a finite normal extension $K \subset N$ such that M can be embedded K -linearly into N . Call this embedding $\beta: M \rightarrow N$. Consider the map ‘precomposing with α ’:

$$f: \text{Emb}_K(M, N) \rightarrow \text{Emb}_K(L, N),$$

so f sends $\beta \in \text{Emb}_K(M, N)$ to $\beta\alpha$. By corollary 2.0.14, since N/K is normal, composing with an automorphism of N over K , we get a surjective map $\text{Aut}_K(N) \rightarrow \text{Emb}_K(M, N)$. Note that this is a transitive action of $G := \text{Aut}_K(N)$ on $\text{Emb}_K(M, N)$. Let $S_1 \subset G$ be the stabilizer of β . Then we get a bijection between G/S_1 and $\text{Emb}_K(M, N)$ compatible with the action of G . Likewise, we have a transitive action of G on $\text{Emb}_K(L, N)$. Let $S_2 \subset G$ be the stabilizer of $\beta\alpha$. Note that $S_1 \subset S_2$. We get a bijection between G/S_2 and $\text{Emb}_K(L, N)$ compatible with the action of G . Moreover, the map f is also compatible with the action of G , so via the two bijections it becomes the natural map $G/S_1 \rightarrow G/S_2$.

This map is visibly surjective with all fibers having the same number of elements. Note that we have an injective map $\text{Emb}_L(M, N) \rightarrow \text{Emb}_K(M, N)$ with image $f^{-1}(\beta\alpha)$, thus $|f^{-1}(\beta\alpha)| = |\text{Emb}_L(M, N)| = [M : L]_S$. The tower law follows from

$$|G/S_1| = |G/S_2| \cdot |f^{-1}(\beta\alpha)|.$$

□

Lemma 3.2.7

Given finite field extension $K \subset L$, $[L : K]_S \leq [L : K]$.

Proof. We can realize $K \subset L$ as a tower of primitive extensions.

$$K \subset K_1 \subset \cdots \subset K_n = L.$$

Then, $K_i = K_{i-1}(\alpha_i)$. We now have that

$$[K(\alpha) : K]_S \leq [K(\alpha) : K].$$

This is because $[K(\alpha) : K]$ denotes the degree and $[K(\alpha) : K]_S$ denotes the number of distinct roots in the splitting field. Then, $[L : K] = \prod [K_i : K_{i-1}]$ and $[L : K]_S = \prod [K_i : K_{i-1}]_S$ and so

$$[L : K]_S \leq [L : K].$$

□

Theorem 3.2.8 (Main Theorem of Separable Extensions)

Let $K \subset L$ be finite field extensions. Then, the following are equivalent:

1. The extension is separable.
2. We can write $L = K(\alpha_1, \dots, \alpha_n)$. Further, suppose that

$$K \subset K(\alpha_1) \subset K(\alpha_1, \alpha_2) \subset \dots \subset K(\alpha_1, \dots, \alpha_n).$$

Then, for each i , $f_i(x) \in K(\alpha_1, \dots, \alpha_{i-1})[x]$, the minimal polynomial of α_i , over $K(\alpha_1, \dots, \alpha_{i-1})$ is separable.

3. $[L : K] = [L : K]_S$.
4. For all towers $K \subset K_1 \subsetneq K_2 \subset L$, there exists a field $K \subset M$ and two distinct elements of $\text{Emb}_{K_1}(K_2, M)$. Equivalently, we will have $[K_2 : K_1]_S \geq 2$.

Proof. (2 $\iff$ 3): ($\implies$) We know that f_i is separable if and only if

$$[K(\alpha_1, \dots, \alpha_i) : K(\alpha_1, \dots, \alpha_{i-1})] = [K(\alpha_1, \dots, \alpha_i) : K(\alpha_1, \dots, \alpha_{i-1})]_S.$$

Then, we have that

$$[L : K] = [K(\alpha_1, \dots, \alpha_i) : K(\alpha_1, \dots, \alpha_{i-1})] \dots [K(\alpha_1, \alpha_2) : K(\alpha_1)][K(\alpha_1) : K].$$

Then, it is apparent that the above is equivalent to

$$[K(\alpha_1, \dots, \alpha_i) : K(\alpha_1, \dots, \alpha_{i-1})]_S \dots [K(\alpha_1, \alpha_2) : K(\alpha_1)]_S [K(\alpha_1) : K]_S.$$

Therefore, $[L : K] = [L : K]_S$.

($\impliedby$): For the reverse, we suppose $[L : K] = [L : K]_S$. Then, we can form the same chain as shown. Then for $K \subset K(\alpha)$, then

$$[K(\alpha) : K]_S \leq [K(\alpha) : K].$$

This is because $[K(\alpha) : K]$ denotes the degree and $[K(\alpha) : K]_S$ denotes the number of distinct roots in the splitting field. However, we have equality of $[L : K] = [L : K]_S$, so we must actually achieve equality: $[K(\alpha) : K]_S = [K(\alpha) : K]$. Therefore, each f_i is separable.

(1 $\implies$ 2): Pick $\alpha_1, \dots, \alpha_n$ such that $L = K(\alpha_1, \dots, \alpha_n)$ and form the tower

$$K \subset K(\alpha_1) \subset K(\alpha_1, \alpha_2) \subset \dots \subset K(\alpha_1, \dots, \alpha_n).$$

We then have that $K(\alpha_1, \dots, \alpha_i) = K(\alpha_1, \dots, \alpha_{i-1})(\alpha_i)$. We know that the minimal polynomial $f_i(x) \in K[x]$ of α_i is separable. Then, consider the minimal polynomial $f_i(X) \in K(\alpha_1, \dots, \alpha_{i-1})[x]$ of α_i . Then, it's clear that $f_i \mid F_i$ in $K(\alpha_1, \dots, \alpha_{i-1})$. Then, F_i being separable implies f_i to be separable.

(3 $\implies$ 1): We have $K \subset L$ and pick any $a \in L$. Then, we want to show that the minimal polynomial $f(x) \in K[x]$ of a is a separable polynomial. We consider $K \subset K(a) \subset L$. Then,

$$\begin{aligned} [L : K] &= [L : K(a)][K(a) : K] \\ [L : K]_S &= [L : K(a)]_S [K(a) : K]_S \end{aligned}$$

Since the LHS of the above are equivalent, we get equality of each of the terms on the RHS. Particularly, $[K(a) : K] = [K(a) : K]_S$, so f is separable.

(3 $\implies$ 4): Suppose we have $K \subset K_1 \subsetneq K_2 \subset L$. Then, we have the following towers:

$$\begin{aligned} [L : K] &= [L : K_2][K_2 : K_1][K_1 : K] \\ [L : K]_S &= [L : K_2]_S[K_2 : K_1]_S[K_1 : K]_S \end{aligned}$$

Thus, the LHS are equivalent again, so we get that each component on the RHS must be equal. Particularly, $[K_2 : K_1] = [K_2 : K_1]_S$. However, $K_2 \neq K_1$, so $[K_2 : K_1] > 1$.

(4 $\implies$ 1): Pick $a \in L$. We want to show that the minimal polynomial $f(x) \in K[x]$ of a is separable. IN characteristic 0, we're done. Now we work in characteristic p . Suppose, for the sake of contradiction, that $f(x)$ is not separable. Then, $f(x) = g(x^p)$, for some irreducible g . Let $b = a^p$. Then, we have

$$K \subset K(b) \subset K(a), \tag{1}$$

where a is a root of $x^p - b \in k(b)[x]$. We now want to show that $K(b) \neq K(a)$. We note that $[K(a) : K] = \deg(f) = p \deg(g) = p[K(b) : K]$. Thus, we know, by the normal tower law, that

$$[K(a) : K(b)] = p.$$

We also know that $x^p - b$ is a minimal polynomial of a over $k(b)$. We also know that $x^p - b = (x - a)^p$, so, viewing (1) as a primitive extension, $[K(a) : K]_S = 1$, which is a contradiction to $[K(a) : K]_S \geq 2$. (Note that this contradicts to our assumption that $f(x)$ is not separable, so $f(x)$ is indeed separable). $\square$

Remark 3.2.9 — We can push the ideas above and prove that

$$[L : K]_S \mid [L : K],$$

and then define the *inseparable degree* $[L : K]_i = \frac{[L : K]}{[L : K]_S}$, and the tower law for $[L : K]_i$. Further, we can show that

$$[L : K]_i = p^{m_i},$$

where p is the characteristic of k . Finally, we say that $K \subset L$ is **purely inseparable** if $[L : K] = [L : K]_i$.

Proposition 3.2.10

We have

$$|\text{Aut}_K(L)| \leq [L : K]_S \leq [L : K]$$

Note that in the above, the first inequality achieves equality if L/K is normal. In the second inequality, we achieve equality if L/K is separable. Note that in the first inequality, if L/K is not normal, then this inequality is strict.

Proof. If L/K is not normal, then we can find an extension $K \subset M$ and two K -linear embeddings $\psi_1, \psi_2 : L \rightarrow M$ such that $\psi_1(L) \neq \psi_2(L)$. Without loss of generality we can

assume that M/K is finite (replace M by the subfield generated over K by the finitely many elements that generate $\psi_1(L)$ and $\psi_2(L)$ over K) and normal (using proposition 2.0.23). The canonical map ‘compose with ψ_1 ’: $\text{Aut}_K(L) \rightarrow \text{Emb}_K(L, M)$ is injective but not surjective because its image does not contain ψ_2 . This gives the desired inequality $\square$

Corollary 3.2.11

Let $K \subset L \subset M$ be a tower of finite field extensions. Then, M/K is separable if and only if M/L and L/K are separable.

Proof. We use the tower laws.

$$[M : L]_S [L : K]_S = [M : K] = [M : K]_S = [M : L]_S [L : K]_S \leq [M : L][L : K].$$

Thus, we must achieve equalities. For the converse, we have

$$[M : K] = [M : L][L : K] = [M : L]_S [L : K]_S = [M : K]_S \leq [M : K].$$

Thus, we must achieve equality at the end. $\square$

§3.3 Perfect Fields

Remark 3.3.1 — I believe this part is nonexaminable.

Definition 3.3.2. A field k is **perfect** if it has characteristic zero, or $\text{char}(k) = p > 0$ and for all $b \in k$, $\exists a \in k$ such that $a^p = b$.

Proposition 3.3.3

A field k is perfect if and only if every irreducible $f(x) \in k[x]$ is separable.

Proof. In characteristic 0, corollary 3.1.4 finishes the argument. Thus, we consider when $\text{char}(k) = p$. ($\implies$): Indeed, suppose $f(x)$ is inseparable. Then, $f(x) = g(x^p)$, where $g(x) \in k[x]$. We write

$$g(x) = b_0 x^n + b_1 x^{n-1} + \cdots + b_n.$$

Let $h = a_0 x^n + a_1 x^{n-1} + \cdots + a_n$, where for each i , we have $a_i^p = b_i$. Then,

$$f(x) = g(x^p) = h(x)^p,$$

since we’re working over characteristic p . This implies $f(x)$ is not irreducible, which is a contradiction.

($\impliedby$): Suppose for the sake of contradiction that there exists an element $b \in k$ such that there’s no $a \in k$ such that $a^p = b$. Then, we claim that $T^p - b$ has only one root in a splitting field; if $\alpha^p = b$, then $T^p - b = (T - \alpha)^p$ by the Freshman’s dream. Further, $T^p - b$ is an irreducible polynomial in $K[T]$. Indeed, every proper polynomial factor of $\alpha^p - b$ is $(T - \alpha)^i$, for some $1 \leq i \leq p - 1$; thus, if this were a factor, then binomial expansion would imply that $i\alpha$ would also be in K . $i \in \mathbb{F}_p^\times$, so $\alpha \in K$ then, which is a contradiction. Thus, $T^p - \alpha$ is irreducible and inseparable in $K[T]$. $\square$

Remark 3.3.4 — This tells us that in characteristic zero, all finite extensions are separable.

Remark 3.3.5 (All finite fields are perfect) — Consider $\mathbb{F}_q$, where $q = p^i$. If we take F such that

$$\begin{aligned}\mathbb{F}_q &\rightarrow \mathbb{F}_q \\ F(a) &= a^p\end{aligned}$$

is a field isomorphism. Thus, all elements have a p -th root.

Given this and the previous discussion about separable extension, we can deduce the following theorem:

Theorem 3.3.6

A field K is perfect if and only if every finite extension of K is a separable extension.

Proof. ($\implies$): Suppose K is perfect: every irreducible polynomial in $K[X]$ is separable. If L/K is a finite extension, then the minimal polynomial in $K[X]$ of every element of L is irreducible by definition; thus, they are separable, so L/K is a separable extension.

($\impliedby$): Now let every finite extension of K be a separable extension. We can just consider primitive extensions, because we can then just induct. Let $f(x) \in K[X]$ be irreducible, and let $L = K(\alpha)$. Then, L/K is a finite separable extension, so by the main theorem of separable extensions, the minimum polynomial of α is separable over K . $\square$

Remark 3.3.7 — This tells us that all finite extensions of finite fields and fields of characteristic zero are separable.

Remark 3.3.8 — This topic will make a return later when we talk about the finite extensions of finite fields (which are Galois).

4 Fundamental Theorem of Galois Theory

Definition 4.0.1. A finite field extension is called a **Galois extension** if it is normal and separable.

Remark 4.0.2 — Therefore, we have that L/K is **Galois** if and only if

$$|\text{Aut}_K(L)| = [L : K]_S = [L : K].$$

Definition 4.0.3. If $K \subset L$ is a Galois extension, then $\text{Aut}_K(L)$ is denoted $\text{Gal}(L/K)$ and is called the **Galois group** of L/K .

Remark 4.0.4 — If $K \subset F \subset L$ are fields and L/K is Galois, then L/F is also Galois. Indeed, by the main theorem about separable extensions, we know that L/F is separable. Corollary 2.0.20 says that L/F is normal. However, F/K is not always Galois.

Example 4.0.5

As an example, take $K = \mathbb{Q}$, $F = \mathbb{Q}(\sqrt[3]{2})$. Then, F/K is not Galois. We can take $L = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$, and this is Galois over $\mathbb{Q}$.

Definition 4.0.6. Let $K \subset L$ be a finite field extension. Let G be a subgroup of $\text{Aut}_K(L)$. Then, define

$$L^G = \{x \in L \mid g(x) = x, \forall g \in G\}.$$

We call $L^G \subset L$ the **fixed field** of G , or the **invariant field** of G .

Theorem 4.0.7 (Artin)

Let L be a field and let $G \subset \text{Aut}(L)$ be a finite subgroup. Then the extension L/L^G is Galois with Galois group $G = \text{Aut}_{L^G}(L)$.

Proof. We claim that $L^G \subset L$ is a finite extension of degree at most $n = |G|$. For this we need to show that any $n + 1$ elements $b_1, \dots, b_{n+1} \in L$ are linearly dependent over L^G . Consider $L[G] = \{ \sum_{g \in G} x_g g \mid x_g \in L \}$. This is a vector space over L of dimension n with basis vectors numbered by the elements $g \in G$. The group G acts on $L[G]$ so that $h \in G$ sends $\sum_{g \in G} x_g g$ to

$$\sum_{g \in G} h(x_g)hg = \sum_{g \in G} h(x_{h^{-1}g})g.$$

Consider the map

$$\phi: L \rightarrow L[G], \quad \phi(b) = \sum_{g \in G} g(b)g.$$

For dimension reasons, $\phi(b_1), \dots, \phi(b_{n+1})$ are linearly dependent over L . Suppose that $m \leq n+1$ is minimal such that we have a linear dependence with non-zero coefficients amongst m of these. Without loss of generality, we reorder the b_i so that the relation is

$$a_1\phi(b_1) + \dots + a_m\phi(b_m) = 0, \quad a_i \in L.$$

Rescaling, we may assume that $a_1 = 1$. Note that $\phi(b_i) \in L[G]$, so the g -coordinate of $\phi(b_i)$ is gb_i . Thus, we rewrite our linear combination as

$$g(b_1) + a_2g(b_2) + \dots + a_mg(b_m) = 0, \quad \forall g \in G.$$

Now, applying $h \in G$, we get also

$$(hg)(b_1) + h(a_2)(hg)(b_2) + \dots + h(a_m)(hg)(b_m) = 0,$$

which is the same thing as saying

$$g'(b_1) + h(a_2)g'(b_2) + \dots + h(a_m)g'(b_m) = 0.$$

This is the g' -coordinate and this holds for every $g' \in G$. We can then rename g' and call it g . We have

$$g(b_1) + a_2g(b_2) + \dots + a_mg(b_m) = g(b_1) + h(a_2)g(b_2) + \dots + h(a_m)g(b_m) = 0.$$

Thus, we have

$$(h(a_2) - a_2)g(b_2) + \dots + (h(a_m) - a_m)g(b_m) = 0,$$

which holds for any $g \in G$. This means that

$$(h(a_2) - a_2)\phi(b_2) + \dots + (h(a_m) - a_m)\phi(b_m) = 0,$$

which contradicts the choice of m unless $h(a_i) = a_i$ for all $i = 2, \dots, m$. But h was any element in G , so $a_i \in L^G$, for $i = 2, \dots, m$. Recall that $a_1 = 1$, so $1 \in L^G$ as well. Therefore, our linear combination

$$a_1\phi(b_1) + \dots + a_m\phi(b_m) = 0$$

has coefficients $a_i \in L^G$. Thus, the $b_1, \dots, b_{n+1}$ are linearly dependent elements of L over L^G : $a_1b_1 + \dots + a_mb_m = 0$ in L . Indeed, G acts trivially on L^G , so we can put the a_i s inside ϕ :

$$\phi(a_1b_1 + \dots + a_mb_m) = 0.$$

We conclude that $[L : L^G] \leq n = |G|$, which is what we wanted to show. For the opposite equality, G is contained in $\text{Aut}_{L^G}(L)$, so $|G| \leq |\text{Aut}_{L^G}(L)|$. Therefore, we get that

$$|G| \leq |\text{Aut}_{L^G}(L)| \leq [L : L^G]_S \leq [L : L^G] \leq |G|,$$

and so L/L^G is Galois and $\text{Gal}(L/L^G) = G = \text{Aut}_{L^G}(G)$. □

Theorem 4.0.8 (Fundamental Theorem)

Let $K \subset L$ be a Galois extension. There is a bijection between the subgroups of $\text{Gal}(L/K)$ and the subfields $K \subset F \subset L$ given by

$$\begin{aligned} G \subset \text{Gal}(L/K) &\mapsto L^G \\ F &\mapsto \text{Aut}_F(L), \end{aligned}$$

where L^G is a subfield of L that contains K . (Both compositions are the identity maps.) Under this correspondence, the normal subgroups $G \subset \text{Gal}(L/K)$ correspond to subextensions $K \subset F \subset L$ such that F/K is Galois. In this case, we have

$$\text{Gal}(F/K) = \text{Gal}(L/K)/\text{Gal}(L/F)$$

Proof. We start with a field F such that $K \subset F \subset L$. We get $\text{Aut}_F(L) \subset \text{Aut}_K(L)$. The fixed field of this subgroup is $L^{\text{Aut}_F(L)}$, call this F_1 . It is apparent that $F \subset F_1$. Then, $\text{Aut}_{F_1}(L) \subset \text{Aut}_F(L)$. However, every $\sigma \in \text{Aut}_F(L)$ acts as the identity on F_1 . Thus, the inclusion $\text{Aut}_{F_1}(L) \subset \text{Aut}_F(L)$ is actually an equality, i.e. $\text{Aut}_{F_1}(L) = \text{Aut}_F(L)$. We know that L is Galois over F and also over F_1 . Thus,

$$[L : F] = |\text{Aut}_F(L)| = |\text{Aut}_{F_1}(L)| = [L : F_1].$$

Hence, $[L : F] = [L : F_1]$, but $F \subset F_1 \subset L$. By the tower law, this implies

$$[L : F] = [L : F_1][F_1 : F],$$

so that $[F_1 : F] = 1$. Hence, $F = F_1$.

Now, let G be a subgroup of $\text{Aut}_K(L) = \text{Gal}(L/K)$. Then, we may use Artin's theorem to conclude that $G = \text{Aut}_{L^G}(L)$.

Recall that proposition 2.0.24 says that if $K \subset F \subset L$ are finite extensions and L/K is Galois and F/K is also Galois, then $\text{Aut}_F(L) = \text{Gal}(L/F)$ is normal in $\text{Gal}(L/K)$ and $\text{Gal}(L/K)/\text{Gal}(L/F) \cong \text{Gal}(F/K)$. It remains to prove the converse that if $G \subset \text{Gal}(L/K)$ is normal, then L^G is normal over K . Note the separability is automatic. Let $i: L^G \hookrightarrow L$ be the given embedding (the tautological embedding). To prove that L^G is normal over K , it's enough to show that if $j: L^G \hookrightarrow L$ is another embedding, then $L^G = i(L^G) = j(L^G)$. By corollary 2.0.14, there is an element $\sigma \in \text{Aut}_K(L) = \text{Gal}(L/K)$ such that $j = \sigma i$. Then,

$$j(L^G) = \sigma i(L^G) = \sigma(L^G).$$

We claim that $\sigma(L^G) = L^{\sigma G \sigma^{-1}}$. Indeed, the elements of $\sigma(L^G)$ are $\sigma(x)$, where $x \in L^G$. The elements of $\sigma G \sigma^{-1}$ are $\sigma g \sigma^{-1}$, where $g \in G$. Then,

$$\sigma g \sigma^{-1}(\sigma(x)) = \sigma g(x) = \sigma(x),$$

because $x \in L^G$. Thus provides that $\sigma(L^G)$ is equivalent to $\sigma G \sigma^{-1}$. However, as G is normal, hence $\sigma G \sigma^{-1} = G$. Thus, $j(L^G) = L^G$. $\square$

Here are some particular cases:

1. If $x \in L$ is fixed by every element of $\text{Gal}(L/K)$, then $x \in K$. Indeed, the Galois correspondence is a bijection. it associates to $K = K \subset L$, where $K = F$. The subgroup $\text{Aut}_K(L)$ inside $\text{Aut}_K(L) = \text{Gal}(L/K)$. Hence, the fixed field of the entire group $\text{Gal}(L/K)$ is K .

2. If $\sigma \in \text{Gal}(L/K)$ fixes every element of L , then $\sigma = e \in \text{Gal}(L/K)$, as $\sigma \in \text{Aut}_L(L) = \{e\}$.

Remark 4.0.9 — The Galois correspondence is inclusion-reversing: bigger subfields correspond to smaller subgroups, and vice versa.

Remark 4.0.10 — The fundamental theorem provides the condition as to when the intermediate field over the base field is Galois.

Exercise 4.0.11. Show that $|\text{Aut}_K(L)|$ divides $[L : K]$.

Solution: We know by Artin's theorem, that if we denote by $G = \text{Aut}_K(L)$, then L/L^G is Galois with Galois group G . Therefore, $|G| = [L : L^G]$. Then, we have that tower law:

$$[L : K] = [L : L^G][L^G : K] = |G|[L^G : K].$$

□

§4.0.i The Compositum

Exercise 4.0.12. Let $K \subset L$ be a finite Galois extension. For intermediate fields $K \subset F_i \subset L$, $i = 1, 2$ define the **compositum** of F_1 and F_2 , denoted as F_1F_2 , to be the smallest subfield of L that contains F_1 and F_2 .

1. Show that $\text{Gal}(L/F_1F_2) = \text{Gal}(L/F_1) \cap \text{Gal}(L/F_2)$.
2. Show that $\text{Gal}(L/(F_1 \cap F_2))$ is the smallest subgroup of $\text{Gal}(L/K)$ that contains $\text{Gal}(L/F_1)$ and $\text{Gal}(L/F_2)$.
3. Show that if F_1 is Galois over K , then F_1F_2 is Galois over F .
4. Show that if F_1 is Galois over K , the elements of $\text{Gal}(F_1F_2/F)$ acting on F_1 defines an isomorphism $\phi: \text{Gal}(F_1F_2/F_2) \rightarrow \text{Gal}(F_1 \cap (F_1 \cap F_2))$.
5. Show that if F_1 is Galois over K , $[F_1F_2 : F_2]$ divides $[F_1 : K]$.

Solution: Solutions by Professor Alexei Skorobogatov:

- (i) The inclusion $\text{Aut}_{F_1F_2}(L) \subset \text{Aut}_{F_1}(L) \cap \text{Aut}_{F_2}(L)$ is clear because $F_i \subset F_1F_2$. Conversely, an automorphism of L that fixes F_1 and F_2 fixes every element of the smallest subfield containing these two fields.
- (ii) The fixed subfield of the smallest subgroup of $\text{Gal}(L/K)$ that contains $\text{Gal}(L/F_1)$ and $\text{Gal}(L/F_2)$ is $F_1 \cap F_2$, so it coincides with $\text{Gal}(L/(F_1 \cap F_2))$ by Galois correspondence.

Remark 4.0.13 — Note that the Galois correspondence is inclusion-reversing, so it reverses the notions “biggest” and “smallest”. In particular, F_1F_2 , being the smallest field containing F_1 and F_2 , corresponds to the biggest subgroup contained in $\text{Gal}(L/F_1)$ and $\text{Gal}(L/F_2)$, and this is their intersection. Likewise, $F_1 \cap F_2$ is the biggest field contained in both F_1 and F_2 , so it corresponds to the smallest subgroup of $\text{Gal}(L/K)$ containing $\text{Gal}(L/F_1)$ and $\text{Gal}(L/F_2)$. Thus (i) and (ii) are formal consequences of the Galois correspondence.

- (iii) Since F_1 is Galois over K , it is a splitting field of some polynomial $f(x) \in K[x]$. Then $f(x)$ of course splits in F_1F_2 , and F_1F_2 is generated by the roots of $f(x)$ over F_2 . Thus F_1F_2/F_2 is normal. Next, F_1/K is separable, hence $F_1 = K(\alpha_1, \dots, \alpha_n)$ such that the minimal polynomial $f_i(x)$ of α_i over the field $K(\alpha_1, \dots, \alpha_{i-1})$ is separable. The minimal polynomial of α_i over $F_2(\alpha_1, \dots, \alpha_{i-1})$ is a factor of $f_i(x)$, so is separable. Thus $F_1F_2 = F_2(\alpha_1, \dots, \alpha_n)$ is separable over F_2 . We conclude that F_1F_2 is Galois over F_2 . (Note that this proof does not require F_2 to be finite over K .) Alternatively, (iii) quickly follows from (i).
- (iv) The kernel fixes both F_1 and F_2 , hence fixes F_1F_2 . Thus ϕ is injective. An element $x \in F_1$ is fixed by $\text{Im}(\phi) \subset \text{Gal}(F_1F_2/F_2)$ if and only if x is fixed by $\text{Gal}(F_1F_2/F_2)$, which happens precisely when $x \in F_1 \cap F_2$. By Galois correspondence we must have $\text{Im}(\phi) = \text{Gal}(F_1/F_1 \cap F_2)$, so ϕ is surjective.
- (v) This follows from (iv) using the tower law for $K \subset F_1 \cap F_2 \subset F_1$.

□

Proposition 4.0.14

Let E and L be extensions of F contained in some common field. If E/F is Galois, then EL/L and $E/E \cap L$ are Galois, and the map

$$\sigma \mapsto \sigma|_E: \text{Gal}(EL/L) \rightarrow \text{Gal}(E/E \cap L)$$

is an isomorphism.

Proof. E/F is Galois, so it is the splitting field of a separable polynomial $f \in F[X]$. As EL is the splitting field of f over L , and E is the splitting field of f over $E \cap L$, we get that EL/L and $E/E \cap L$ are Galois. Every automorphism σ of EL fixing the elements of L maps roots of f to roots of f , and so $\sigma E = E$. Thus, we have a homomorphism as defined in the proposition statement. If $\sigma \in \text{Gal}(EL/L)$ fixes the elements of E then it fixes the elements of EL , and hence is the identity map. Therefore, $\sigma \mapsto \sigma|_E$ is injective. If $\alpha \in E$ is fixed by all $\sigma \in \text{Gal}(EL/L)$, then $\alpha \in E \cap L$. By Artin's theorem, this implies that the image of $\sigma \mapsto \sigma|_E$ is $\text{Gal}(E/E \cap L)$. □

5 Galois Group of Polynomials

§5.1 Galois Groups

Definition 5.1.1. Let k be a field and let $f(x) \in k[x]$ be a separable polynomial. Let L be a splitting field of $f(x)$. The **Galois group** of $f(x)$ is $\text{Gal}(L/k)$. We denote it by $\text{Gal}(f)$.

Remark 5.1.2 — Note that in the above definition, L is a Galois extension of k by the main theorem of normal extensions.

Remark 5.1.3 — Note that all splitting fields of $f(x)$ are isomorphic over k , so $\text{Gal}(L/k)$ doesn't depend on L up to isomorphism.

Let $n = \deg(f(x))$. Let $\alpha_1, \dots, \alpha_n$ be the roots of $f(x)$ in L . Assuming $f(x)$ is monic, we have $f(x) = \prod_i (x - \alpha_i)$. We have, for any $\sigma \in \text{Gal}(L/k)$ that σ acts on $\{\alpha_1, \dots, \alpha_n\}$ as a permutation. This can be seen as below.

$$f(x) = x^n + \dots + a_1x + a_0,$$

then

$$0 = f(\alpha_i) = \alpha_i^n + \dots + a_1\alpha_i + a_0$$

implies that

$$0 = \sigma(f(\alpha_i)) = \sigma(\alpha_i)^n + \dots + a_1\sigma(\alpha_i) + a_0,$$

and so $\sigma(\alpha_i)$ is a root of $f(x)$ as well. Therefore, we get a group homomorphism $\text{Gal}(L/k) = \text{Gal}(f) \rightarrow S_n$, where S_n is the permutation group on n letters. This is a homomorphism, because on both sides, the group law is composition. This homomorphism $\text{Gal}(f) \rightarrow S_n$ is injective. Indeed, suppose $\sigma \in \text{Gal}(f) = \text{Gal}(L/k)$ fixes $\alpha_1, \dots, \alpha_n$. However, $L = k(\alpha_1, \dots, \alpha_n)$. Hence, σ acts trivially on L , and so $\sigma = e \in \text{Gal}(f)$.

We introduce some philosophy here. For any separable polynomial $f(x)$, we defined a subgroup of S_n , where $n = \deg(f)$.

Example 5.1.4

Let the $\text{char}(k) \neq 2$ and consider $f(x) = x^2 - a$ for nonzero a . Then,

- $\text{Gal}(f) \cong \mathbb{Z}/2$ if a is not a square in k . Here, we have $L = k(\sqrt{a})$.
- $\text{Gal}(f) = \{e\}$ if $a = b^2$, for some $0 \neq b \in k$. Here, $L = k$.

Proposition 5.1.5

The action of $\text{Gal}(f)$ on the roots of $f(x)$ in L defines an injective homomorphism $\text{Gal}(f) \hookrightarrow S_n$, where $n = \deg(f(x))$. The orbits of the action of $\text{Gal}(f)$ on the roots bijectively correspond to the roots of irreducible factors of $f(x)$. In particular, if $f(x)$ is irreducible, then the action is transitive, i.e. there is only one orbit.

Proof. Let $L = k(\alpha_1, \dots, \alpha_n)$, where $f(x) = \prod_{i=1}^n (x - \alpha_i)$. Thus, if $g \in \text{Gal}(f)$, $g\alpha_i = \alpha_i$ for all i , then $g = e \in \text{Gal}(f) = \text{Aut}_k(L)$. Now let $g(x)$ be an irreducible factor of $f(x)$, say

$$g(x) = \prod_{i=1}^m (x - \beta_i).$$

We must show that for any $i = 2, \dots, m$, there is a $g \in \text{Gal}(f)$ such that $g(\beta_1) = \beta_i$. We have $K := k[x]/(g(x)) \cong k(\beta_1) \cong k(\beta_i)$, so we have embeddings of $K \hookrightarrow L$ such that the image of x in K is sent to β_1 and another embedding sends it to β_i . By corollary 2.0.14, since L/k is normal, these embeddings differ by an automorphism of L over k . Therefore, $\exists g \in \text{Gal}(f)$ such that $\beta_i = g(\beta_1)$. $\square$

Exercise 5.1.6. Let L be a normal extension of K . Consider the action of the group $G := \text{Aut}_K(L)$ on L . Prove that the orbits $G \cdot b$ are precisely the sets of roots of irreducible polynomials $f \in K[x]$ which split into linear factors in L . That is, if f is the minimal polynomial of b over K , then the roots of f are precisely the set $G \cdot b$.

As a consequence, show that $\text{Aut}_K(L)$ acts transitively on the set of roots of every irreducible polynomial $f \in K[x]$ which splits into linear factors in L .

Solution. It is clear that G preserves the set of roots of the irreducible polynomial $f \in K[x]$. On the other hand, $\prod_{\beta \in G \cdot b} (x - \beta)$ is preserved under G , so is in $K[x]$. However, this also divides f and has b as a root, so this polynomial must equal f . That is, $G \cdot b$ contains all the roots of f . $\square$

§5.2 Transitive Subgroups of S_n

Given the previous proposition, we understand that we need to study the subgroups of S_n and question which ones are transitive.

We consider low degree (separable) polynomials $f(t)$ at the start:

§5.2.i Degree 2

Let $f(t) = t^2 + bt + c = 0$, where $b, c \in k$. If $f(t)$ has a root in k , then $\text{Gal}(f) = \{e\}$. Otherwise, $\text{Gal}(f) \cong C_2$.

§5.2.ii Degree 3

If $f(x)$ is completely split in k , then $\text{Gal}(f) = \{e\}$.

If $f(x)$ is a product of linear polynomials and an irreducible quadratic, then $\text{Gal}(f) \cong C_2$.

If $f(x)$ is irreducible, then $\text{Gal}(f) \cong C_3$ or $\text{Gal}(f) \cong S_3$, but we need some criterion to distinguish these two cases.

§5.2.iii The Vandermonde Determinant

For this, we use the Vandermonde determinant. Let $f(x) = (x - \alpha_1) \dots (x - \alpha_n)$. Then, recall that the **Vandermonde determinant** is the following determinant:

$$\det \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_n^2 \\ \vdots & \vdots & \dots & \vdots \\ \alpha_1^{n-1} & \alpha_2^{n-1} & \dots & \alpha_n^{n-1} \end{pmatrix} = \prod_{i < j} (\alpha_j - \alpha_i)$$

Via linear algebra, swapping two columns multiplies this by -1 . If $\sigma \in S_n$, then $\text{sign}(\sigma)$ is defined as the sign by which this determinant is multiplied when we permute the columns by σ . Note that the group A_n consists of all permutation $\sigma \in S_n$ such that $\text{sign}(\sigma) = 1$.

Definition 5.2.1. The **discriminant** of $f(x)$ is $\Delta_f := \prod_{i < j} (\alpha_i - \alpha_j)^2$.

Clearly, $\Delta_f \neq 0$ if and only if $f(x)$ is separable. Suppose $f(x)$ is separable. Then $\text{Aut}_k(L) = \text{Gal}(f) \subset S_n$ preserves Δ_f , so by the fundamental theorem, $\Delta_f \in L^{\text{Gal}(f)} = k$. Let $\sqrt{\Delta_f} = \pm \prod_{i < j} (\alpha_i - \alpha_j) \in L$. Thus, we have $k \subset k(\sqrt{\Delta_f}) \subset L$.

Proposition 5.2.2

Assume $\text{char}(k) \neq 2$ and $f(x) \in k[x]$ is separable. Then, $\text{Gal}(f) \subset A_n$ if and only if Δ_f is a square in k (i.e. $\sqrt{\Delta_f} \in k$)

Proof. We have

$$\sqrt{\Delta_f} = \pm \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_n^2 \\ \vdots & \vdots & \dots & \vdots \\ \alpha_1^{n-1} & \alpha_2^{n-1} & \dots & \alpha_n^{n-1} \end{pmatrix}.$$

Hence, $\sigma(\sqrt{\Delta_f}) = \text{sign}(\sigma) \cdot \sqrt{\Delta_f}$. $\text{char}(k) \neq 2$, so we have $-1 \neq 1$, so an element $\sigma \in \text{Gal}(f)$ fixes $\sqrt{\Delta_f}$ if and only if $\sigma \in A_n$. In other words, $\sqrt{\Delta_f} \in L^{\text{Gal}(f)} \iff \text{Gal}(f) \subset A_n$. Note that $L^{\text{Gal}(f)} = k$. $\square$

Remark 5.2.3 — Consider when $n = 2$. Then $\Delta_f = (\alpha_1 - \alpha_2)^2 = b^2 - 4c$, where $f(t) = t^2 + bt + c$ (the usual discriminant).

Remark 5.2.4 — By the above, we get that when Δ_f is a square, the Galois group $\text{Gal}(f) \cong C_2 \cong A_2$; if Δ_f is a nonsquare, it follows that $\text{Gal}(f) \cong S_2$.

Formula for Δ_f

We want a formula for Δ_f . Note that $\Delta_f = 0 \iff f$ is not separable $\iff f(x)$ and $f'(x)$ have a common root.

Definition 5.2.5. Let $f(x) = a \prod_{i=1}^n (x - \alpha_i)$ and $g(x) = b \prod_{i=1}^m (x - \beta_i)$, where $a, b \neq 0$, and $\alpha_i, \beta_j \in L$, where L is a splitting field of $f(x)g(x)$. The **resultant** of $f(x)$ and $g(x)$ is

$$\text{Res}(f, g) := a^m b^n \prod_{i,j} (\alpha_i - \beta_j).$$

Proposition 5.2.6

Let $f(x)$ and $g(x)$ be as above. Then,

$$\text{Res}(f, g) = (-1)^{mn} \text{Res}(g, f) = a^m \prod_{i=1}^n g(\alpha_i).$$

Consequently,

$$\text{Res}(f, g) = 0 \iff f(x) \text{ and } g(x) \text{ have a common root.}$$

If at least one of $f(x)$ and $g(x)$ is separable, then $\text{Res}(f, g) \in k$.

Proof. The formulae is apparent by definition. Let $K = k(\alpha_1, \dots, \alpha_n) \subset L$, where K is a splitting field of $f(x)$. As $f(x)$ is separable, K/k is Galois. The elements of $\text{Gal}(f)$ permute $\alpha_1, \dots, \alpha_n$, hence they permute $g(\alpha_1), \dots, g(\alpha_n)$. Thus, $\prod_{i=1}^n g(\alpha_i) \in K^{\text{Gal}(f)} = k$, by the fundamental theorem. $\square$

Proposition 5.2.7

Let $f(x) \in k[x]$ be a monic polynomial with derivative $f'(x)$. Then

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \text{Res}(f, f') = (-1)^{\frac{n(n-1)}{2}} \text{Res}(f', f)$$

Proof. This is by calculation.

$$f'(x) = \sum_{i=1}^n \prod_{j \neq i} (x - \alpha_j)$$

Thus,

$$\text{Res}(f, f') = \prod_{i=1}^n f'(\alpha_i) = \prod_{i=1}^n \prod_{j \neq i} (\alpha_i - \alpha_j) = (-1)^{\frac{n(n-1)}{2}} \prod_{i < j} (\alpha_i - \alpha_j)^2 = (-1)^{\frac{n(n-1)}{2}} \Delta_f.$$

$\square$

Now, how do we calculate $\text{Res}(f, g)$? If $g(x) = h(x)f(x) + r(x)$, then $g(\alpha_i) = 0 + r(\alpha_i)$, where α_i is a root of $f(x)$. Therefore,

$$\text{Res}(f, g) = a^{m-m_1} \text{Res}(f, r),$$

where $m_1 = \deg(r(x))$ and $m = \deg(g(x))$.

Exercise 5.2.8. Let $f(x) = x^n + ax + b$, where $a, b \in k$. Prove that

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \left((-1)^{n-1} (n-1)^{n-1} a^n + n^n b^{n-1} \right).$$

Example 5.2.9

In particular, if $n = 3$ and $g(x) = x^3 + px + q$, then $\Delta_f = -4p^3 - 27q^2$.

Remark 5.2.10 — This is the general form of cubic polynomials when $\text{char}(k) \neq 3$.

Now we answer the original question: An irreducible cubic has Galois group C_3 if $\Delta_f \in k^2$. Otherwise, $\text{Gal}(f) \cong S_3$.

§5.2.iv Degree 4

Let us now return to our analysis on low degree polynomials. We want to find the subgroups of S_4 first. Up to conjugation the elements of S_n are given by their cycle shape. In S_4 , we have the following cycle shapes: e , transpositions (2-cycle), 3-cycle, 4-cycle, double transpositions (product of two independent 2-cycles). We can then split our cases. Note that the non-transitive cases are easy, so we try to find transitive subgroups of S_4 . We have:

- C_4 , which is generated by a 4-cycle. $\text{sign}((1234)) = -1$, so $C_4 \not\subset A_4$ and is not normal.
- $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$. This is a normal subgroup, as the elements $(12)(34), (13)(24), (14)(23)$ form one conjugacy class. $V_4 \subset A_4$.
- D_8 , the Dihedral group of order 8. It is generated by (1234) and (13) . It's not normal and not contained in A_4 .
- A_4 is normal in S_4 , as it has index 2.
- S_4 .

Remark 5.2.11 — We make a remark on the Klein-four group. S_4 acts on $V_4 \setminus \{e\}$ by conjugation. This provides a surjective homomorphism $S_4 \rightarrow S_3$. Note that $V_4 \cong C_2 \times C_2$. Therefore, $S_4/V_4 \cong S_3$.

Unfortunately, Δ_f is not enough to determine $\text{Gal}(f)$ in this case. $\Delta_f \in k^2 \iff \text{Gal}(f)$ is V_4 or A_4 , and $\Delta_f \notin k^2 \iff \text{Gal}(f)$ is C_4 , D_4 or S_4 .

Definition 5.2.12. Let $f(x) \in k[x]$ be a separable monic polynomial of degree 4 with a splitting field L . Write

$$f(x) = \prod_{i=1}^4 (x - \alpha_i).$$

Define

$$a = \alpha_1\alpha_2 + \alpha_3\alpha_4, \quad b = \alpha_1\alpha_3 + \alpha_2\alpha_4, \quad c = \alpha_1\alpha_4 + \alpha_2\alpha_3.$$

The polynomial $g(x) = (x - a)(x - b)(x - c)$ is called the **resolvent cubic** of $f(x)$.

Proposition 5.2.13

The resolvent cubic is separable and $\Delta_f = \Delta_g$. Assuming $\text{char}(k) \neq 2$, the fields

$$k \subset k(\sqrt{\Delta_f}) \subset K = k(a, b, c) \subset L$$

are the fixed subfields of the normal subgroups of

$$\text{Gal}(f) \supset \text{Gal}(f) \cap A_4 \supset \text{Gal}(f) \cap V_4 \supset \{e\}.$$

Moreover, $g(x) \in k[x]$.

Proof. First note $a - b = (\alpha_1 - \alpha_4)(\alpha_2 - \alpha_3)$, $b - c = (\alpha_1 - \alpha_2)(\alpha_3 - \alpha_4)$, $a - c = (\alpha_1 - \alpha_3)(\alpha_2 - \alpha_4)$. Then, $\Delta_f = \Delta_g$. Note that $f(x)$ being separable implies that $g(x)$ is separable. Now let $\sigma \in \text{Gal}(f) \subset S_4$. Then, $\sigma(\sqrt{\Delta_f}) = \text{sign}(\sigma) \cdot \sqrt{\Delta_f}$. Thus, $\sqrt{\Delta_f}$ is fixed by every $\sigma \in \text{Gal}(f)$ if and only if $\text{Gal}(f) \subset A_4$. We have $k(\sqrt{\Delta_f}) = L^{\text{Gal}(f) \cap A_4}$. Equivalently, $\text{Gal}(f) \cap A_4 = \text{Aut}_{k(\sqrt{\Delta_f})}(L) = \text{Gal}(L/k(\sqrt{\Delta_f}))$.

Next, if $\sigma \in \text{Gal}(f) \cap V_4$, then σ fixes each of a, b, c , as V_4 contains double transpositions. Hence, fixes every element of $K = k(a, b, c)$. S_4 permutes $\alpha_1, \alpha_2, \alpha_3, \alpha_4$, hence permutes a, b, c . The stabilizer of $b = \alpha_1\alpha_3 + \alpha_2\alpha_4$ is the subgroup of S_4 generated by $\langle (1234), (13) \rangle \cong D_8$. (1234) sends $\alpha_1\alpha_3$ to $\alpha_2\alpha_4$ and sends $\alpha_2\alpha_4$ to $\alpha_1\alpha_3$. (13) fixes each part. Note that the action of S_4 on $\{a, b, c\}$ is transitive, so stabilizers of points are all conjugate. A permutation $\sigma \in S_4$ fixes each of a, b, c if and only if σ belongs to the intersection of the stabilizers of a, b, c . Equivalently, $\sigma \in \bigcap_{g \in S_4} gD_8g^{-1}$. The claim is that this intersection is V_4 . Clearly, $V_4 \subset D_8$ and V_4 is normal in S_4 . If $\sigma \in \text{Gal}(f) \subset S_4$, then if σ fixes each of a, b, c , then $\sigma \in V_4$. Thus, $\text{Gal}(L/K) = \text{Aut}_K(L) = \text{Gal}(f) \cap V_4$.

Finally, $\text{Gal}(f)$ permutes $\{a, b, c\}$, hence acts trivially on the coefficients of $g(x)$. By the fundamental theorem, this implies that $g(x) \in k[x]$. $\square$

Remark 5.2.14 — Note that in the above, we have

$$\begin{array}{ccccccc} k & & k(\sqrt{\Delta_f}) & & K = k(a, b, c) & & L \\ \updownarrow & & \updownarrow & & \updownarrow & & \updownarrow \\ \text{Gal}(f) & & \text{Gal}(f) \cap A_4 & & \text{Gal}(f) \cap V_4 & & \{e\} \end{array}$$

Solving Quartics by Radicals

Suppose $\text{char}(k) \neq 2$. We can reduce any quartic to

$$f(x) = x^4 + px^2 + qx + r, \quad p, q, r \in k.$$

Proposition 5.2.15

The resolvent cubic of $f(x)$ as above is

$$g(x) = x^3 - px^2 - 4rx + (4rp - q^2).$$

Proof. This is a direct calculation. □

Algorithm for Determining $\text{Gal}(f)$

Let $f(x)$ be irreducible.

1. $\Delta_f = \Delta_g$ is a square in k , then $\text{Gal}(f) \subset A_4$:

- a) $\text{Gal}(f) = A_4$ if $g(x)$ has no roots in k (irreducible).
- b) $\text{Gal}(f) = V_4$ if $g(x)$ has a root in k (completely split).

Indeed, $\text{Gal}(g) = \text{Gal}(f)/\text{Gal}(f) \cap V_4$, by proposition 5.2.13. $A_4/V_4 \cong C_3$, and $V_4/V_4 = \{e\}$.

2. If $\Delta_f = \Delta_g$ is not a square in k , then $\text{Gal}(f)$ is not contained in A_4 :

- a) If $g(x)$ has a root in k (exactly one), then $\text{Gal}(f) = D_8$ if f remains irreducible over K .
- b) If $g(x)$ has a root in k (exactly one), then $\text{Gal}(f) = C_4$, if f is a product of two quadratics over K .
- c) If $g(x)$ has no roots in k , then $\text{Gal}(f) = S_4$.

The action of S_4 on $\{a, b, c\}$ is via $S_4/V_4 \cong S_3$. $D_8/V_4 \cong C_2 \subset S_3$. Also, $C_4/V_4 \cap V_4 \cong C_2 \subset S_3$. The Galois group of f over K is $\text{Gal}(f) \cap V_4$ by proposition 5.2.13. Note that this intersection is V_4 if $\text{Gal}(f) = D_8$ and C_2 if $\text{Gal}(f) = C_4$.

Now, how did the resolvent cubic appear *historically*? We introduce a new variable t , and we rewrite

$$f(x) = x^4 + px^2 + qx + r$$

as

$$f(x) = \left(x^2 + \frac{t}{2}\right)^2 - \left((t-p)x^2 - qx + \left(\frac{t^2}{4} - r\right)\right).$$

The second quadratic has discriminant zero precisely when $g(t) = 0$:

$$q^2 - (t-p)(t^2 - 4r) = 0$$

This gives a practical way to solve the original quartic via rationals.

§5.3 Biquadratic Extensions

Here, we ask what a quadratic extension of a quadratic extension is.

Remark 5.3.1 — We assume that all fields have characteristic not equal to 2.

The following lemma allows us to distinguish between different quadratic extensions.

Lemma 5.3.2

Let K be a field. Let $a, b \in K^\times$.

1. If a is not a square in K , then the kernel of the natural map

$$K^\times / K^{\times 2} \rightarrow K(\sqrt{a})^\times / K(\sqrt{a})^{\times 2}$$

consists of the classes of 1 and a .

2. We have $K(\sqrt{a}) = K(\sqrt{b})$ if and only if the class of ab in $K^\times / K^{\times 2}$ is trivial (1) if and only if the class of $\frac{a}{b}$ is trivial.
3. We have $K(\sqrt{a}, \sqrt{b}) = K(\sqrt{a})$ if and only if either b or ab is a square in K . If $K(\sqrt{a}, \sqrt{b}) \neq K(\sqrt{a}) \neq K$, then $\text{Gal}(K(\sqrt{a}, \sqrt{b})/K) \cong C_2 \times C_2$.
4. Assume that b is not a square in K . If $K(\sqrt{x + y\sqrt{b}}) = K(\sqrt{b})$, where $x, y \in K$, then the norm $N(x + y\sqrt{b}) = x^2 - by^2$ is a square in K .

Proof. (1): We have $(z + w\sqrt{a})^2 = z^2 + aw^2 + 2zw\sqrt{a} \in K$. If $z = 0$, then we have aw^2 . $a \in K^{\times 2}$, which is a contradiction. Thus, $w = 0$, and so our elements is z^2 , so it's in $K^{\times 2}$.

(2): If two fields are equal, then their kernels are equal. Then, a gives rise to the same class $K^\times / K^{\times 2}$ as b . Thus, the class of the product is 1. For the converse, if the class of ab is trivial (equivalently if the class of $\frac{a}{b}$ is trivial), then a is b multiplied by a square, in which case the two extensions are equivalent.

(3): $(K(\sqrt{a}, \sqrt{b}) = K(\sqrt{a})$ implies that b is a square in $K(\sqrt{a})$. This implies that either $b \in K^{\times 2}$ or $b = aw^2$, where $w \in K^\times$. For the converse, if b is a square, then $\sqrt{b}$ doesn't do much. If ab is a square, then we have $\sqrt{a}\sqrt{b}$ times some element, so it's not entirely difficult as well.

(4): $K(\sqrt{x + y\sqrt{b}}) = K(\sqrt{b})$ is equivalent to when $x + y\sqrt{b}$ is a square in $K(\sqrt{b})$. This is equivalent to saying

$$x + y\sqrt{b} = (z + w\sqrt{b})^2.$$

This implies that $x - y\sqrt{b} = (z - w\sqrt{b})^2$. This implies that

$$x^2 - by^2 = (z^2 - bw^2) \in K^{\times 2}.$$

□

Now let K be a field and let $a, b \in K$. We will study the field extension

$$L = K(\sqrt{a \pm \sqrt{b}}) = K(\sqrt{a + \sqrt{b}}, \sqrt{a - \sqrt{b}}) \quad (*)$$

Assume that b is not a square in K , otherwise L is not too interesting. In particular, $b \neq 0$. Define $c = a^2 - b$. Then, $K(\sqrt{c}) = L$. We make the following observations.

1. The field L is the splitting field of the polynomial

$$f(x) = (x^2 - a)^2 - b = x^4 - 2ax^2 + c.$$

2. The extension L/K is separable hence Galois. This is because $f' = 4x^3 - 4ax$, and since $\text{char}(K) \neq 2$, its roots are $0, \pm\sqrt{a}$, which are not roots of f .
3. $f(x)$ has no roots in K (because $\sqrt{b} \notin K$).
4. If b and c are not squares, then $f(x)$ is irreducible.

Proof. Suppose that $f(x)$ is reducible. Then, either $\pm a \pm \sqrt{b} \in K$ or $\pm\sqrt{a^2 - b} = \pm\sqrt{c} \in K$, which are both contradictions. $\square$

5. Consider the following identity:

$$\left(\sqrt{a + \sqrt{b}} + \sqrt{a - \sqrt{b}}\right)^2 = 2a + 2\sqrt{a^2 - b} = 2a + \sqrt{c}.$$

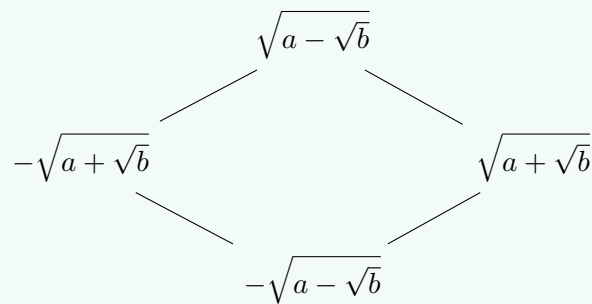
Hence, $K(\sqrt{c}) \subset L$ and L also contains the elements $\sqrt{2(a \pm \sqrt{c})}$. Note that if we iterate this construction we get back $2\sqrt{a \pm \sqrt{b}}$, so this shows that $L = K(\sqrt{2(a \pm \sqrt{c})})$. In other words, L is also the splitting field of the *companion polynomial*

$$\begin{aligned} g(x) &= \left(x - \sqrt{2(a + \sqrt{c})}\right) \left(x + \sqrt{2(a + \sqrt{c})}\right) \left(x - \sqrt{2(a - \sqrt{c})}\right) \left(x + \sqrt{2(a - \sqrt{c})}\right) \\ &= x^4 - 4ax^2 + 4b, \end{aligned}$$

and the companion polynomial of g is $16f\left(\frac{x}{2}\right)$.

Proposition 5.3.3

Every automorphism of L preserves the following square (i.e., acts by rotation or reflection on it):



In particular, the injection $\text{Gal}(L/K) \hookrightarrow S_4$ has image inside D_8 .

Proof. It is enough to show that every automorphism sends opposite vertices to opposite vertices. This is clear because opposite vertices are negatives. $\square$

Theorem 5.3.4

Let K be a field of $\text{char}(K) \neq 2$. Let $a, b \in K^\times$ with b not a square. Consider the Galois extension L/K given by $(*)$ and set $c := a^2 - b$. Then there is an injection $\text{Gal}(L/K) \hookrightarrow D_8$, and we have the following:

1. If bc, c are not squares in K , then $[L : K] = 8$ so that $\text{Gal}(L/K) \cong D_8$.
2. If bc is a square in K , then $[L : K] = 4$ and $\text{Gal}(L/K) \cong C_4$.
3. If c is a square in K , then there are the following two cases:
 - a) If neither $2(a + \sqrt{c})$ nor $2(a - \sqrt{c})$ is a square in K , then $[L : K] = 4$ and $\text{Gal}(L/K) \cong V_4 \cong C_2 \times C_2$ (the action is transitive).
 - b) If one of $2(a \pm \sqrt{c})$ is a square, then $L = K(\sqrt{b})$ and $\text{Gal}(L/K) \cong C_2$.

Proof. We can use the theory of quartic polynomials. The resolvent cubic of $f(x)$ is

$$t^3 + 2at^2 - 4ct - 8ac = (t + 2a)(t^2 - 4c),$$

so it has a root in K . Up to a square in K , the discriminant is c . Moreover, $K(\sqrt{c})$ is a splitting field of the resolvent cubic. If c is not a square in K , then $\text{Gal}(L/K)$ is D_8 or C_4 . If c is a square in K , then $\text{Gal}(L/K) \subset V_4$, with equality if $f(x)$ is irreducible.

3. Assume that c is a square in K . Then $L = K(\sqrt{2(a + \sqrt{c})}, \sqrt{2(a - \sqrt{c})})$. The product $2(a + \sqrt{c}) \cdot 2(a - \sqrt{c}) = 4b$ is not a square, so we are either in case 1. or in case 2. The statement follows from the third statement in lemma 5.3.2.

1. Assume that neither of b, c, bc is a square in K . Then $[K(\sqrt{b}) : K] = 2$. We have $[K(\sqrt{a + \sqrt{b}}) : K(\sqrt{b})] = 2$, otherwise $c = a^2 - b$ is a square in K by the fourth statement in lemma 5.3.2. Since $(a + \sqrt{b})(a - \sqrt{b}) = c$ is not a square in $K(\sqrt{b})$, the fields $K(\sqrt{a + \sqrt{b}})$ and $K(\sqrt{a - \sqrt{b}})$ are two distinct quadratic extensions of $K(\sqrt{b})$, so L has degree 2 over $K(\sqrt{a + \sqrt{b}})$. The tower law implies $|\text{Gal}(L/K)| = [L : K] = 8$, so by proposition 5.3.3 we have $\text{Gal}_K(L) \cong D_8$. (Alternatively, one can prove that all subgroups of S_4 of order 8 are conjugate to D_8 .)

2. Assume that bc is a square in K . Now c is not a square in K and $\sqrt{b} \in K(\sqrt{c})$. Thus $f(x) = (x^2 - (a + \sqrt{b}))(x^2 - (a - \sqrt{b}))$ splits into two irreducible quadratics over $K(\sqrt{c})$, so $\text{Gal}(L/K) \cong C_4$. $\square$

Remark 5.3.5 — The proof shows that $f(x)$ is irreducible in cases 1, 2, 3a), but not in the case 3b).

Example 5.3.6

If K is the splitting field of $x^4 - 2$ over $\mathbb{Q}$, then $a = 0$, $c = -2$, $b = 2$, $bc = -4$, so $G = D_4$ and we get the full Hasse diagram of 10 fields.

Example 5.3.7

If K is the splitting field of $x^4 - 4x^2 + 2$ over $\mathbb{Q}$, then $a = 2$, $c = 2$, $b = 2$, $bc = 4$, so we get $G = C_4$ and $K = \mathbb{Q}(\sqrt{2 + \sqrt{2}})$.

Example 5.3.8

If K is the splitting field of the polynomial $x^4 - 10x^2 + 1$ over $\mathbb{Q}$ (here $a = 5$, $c = 1$, $b = 24$, $bc = 24$, $2(a \pm \sqrt{c}) = \{8, 12\}$), then $G = C_2 \times C_2$. In this case, $K = \mathbb{Q}(\sqrt{2 + \sqrt{3}})$, which explains the identity

$$\sqrt{5 + 2\sqrt{6}} = \sqrt{2} + \sqrt{3}.$$

At least, Galois theory implies that the LHS must be of the form $A\sqrt{2} + B\sqrt{3}$ for some $A, B \in \mathbb{Q}$.

6 More Topics

§6.1 Frobenius Lifting

The aim of this section is to calculate the Galois group of a polynomial (of arbitrary degree) with integer coefficients.

§6.1.i Finite Fields

Remark 6.1.1 — Here, we will let p be prime.

Let K be a finite field. We recall the classification of finite fields. Then, there is exactly one homomorphism $\phi: \mathbb{Z} \rightarrow K$ such that $\phi(1) = 1$, $|F| < \infty$, with $\ker(\phi) \neq 0$. There are no zero divisors in K , so $\ker(\phi)$ is a prime ideal. Thus, there exists a prime $p > 0$ such that $\ker(\phi) = p\mathbb{Z}$ and we call this unique prime the characteristic of K , denoted $\text{char}(K) = p$. Thus, $\text{Im}(\phi) = \mathbb{Z}/p = \mathbb{F}_p$ is a subfield of K . Therefore, K is a finite dimensional vector space over $\mathbb{F}_p$. Therefore, $|K| = p^d$, where $d \geq 1$.

Proposition 6.1.2

There is a unique finite field with p^d elements.

Proof. All fields of size p^d are isomorphic. Indeed, $K^\times$ is a cyclic group of order $p^d - 1$. For any $x \in K$ (including 0), it is a root of

$$x^{p^d} - x = x(x^{p^d-1} - 1).$$

Since this has at most p^d roots, there are exactly this many roots. Therefore, these roots are precisely all elements of K (without repetitions). Therefore, K is a splitting field of $x^{p^d} - x$. Note that all splitting fields of a given polynomial are isomorphic. Consequently, there is a unique field with p^d elements, up to isomorphism.

Conversely, take $x^{p^d} - x$. A splitting field of this polynomial exists. □

Exercise 6.1.3. Let K be a finite field of characteristic p , with size $q = p^r$. Show that a field L of size p^s is a subfield of K if and only if $s \mid r$.

Proof. ($\implies$): Let $F = \mathbb{F}_p$. Then, this follows by the tower law:

$$r = [K : F] = [K : L][L : F] = [K : L] \cdot s.$$

($\impliedby$): This follows since the roots of the polynomial $X^{p^s} - X$ (which provide the elements of L) are also roots of the polynomial $X^{p^r} - X$ (which provide the elements of K). Thus, the elements of L are elements of K , proving inclusion. □

Definition 6.1.4. Let $q = p^d$, $d \geq 1$. Define $\mathbb{F}_q$ as a field of size $q = p^d$.

Definition 6.1.5. Let K be any field of characteristic p . Then, the **Frobenius map** $\text{Frob}_p: K \rightarrow K$ is given by

$$\text{Frob}_p(x) = x^p.$$

We also define $\text{Frob}_{p^d} = (\text{Frob}_p)^d$ to be the map $x \mapsto x^{p^d}$.

Remark 6.1.6 — For all $x, y \in K$ with $\text{char}(K) = p$, we have:

$$\begin{aligned}(x + y)^p &= x^p + y^p \\ (xy)^p &= x^p y^p\end{aligned}$$

Thus, Frob_p is a field homomorphism. This tells us that Frob_p is always injective, but we can see this directly as well (next remark).

Remark 6.1.7 — Frob_p is injective:

$$x^p = 0 \implies x = 0,$$

since we don't have zero-divisors.

However, Frob_p is not always surjective (see next example).

Example 6.1.8

Let $K = \mathbb{F}_p(x) = \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in \mathbb{F}_p[x], g \neq 0 \right\}$. Then

$$(a_n x^n + \cdots + a_0)^p = a_n^p (x^p)^n + \cdots + a_1^p x^p + a_0^p, \quad a_i \in \mathbb{F}_p.$$

By Fermat's little theorem, we have this is equal to

$$a_n (x^p)^n + \cdots + a_1 x^p + a_0.$$

Therefore, the image of $\text{Frob}_p(K) = \mathbb{F}_p(x^p) \subsetneq \mathbb{F}_p(x)$.

Proposition 6.1.9

Let K be a finite field with $q = p^d$ elements. Then, $\text{Frob}_p(K) = K$. Hence, Frob_p is an automorphism of K . The order of Frob_p is $d = [K : \mathbb{F}_p]$.

Proof. A map from a finite set to itself, which is injective, is also surjective. Since $|K| = p^d$, every element of K is a root of $x^{p^d} - x$. Hence, $(\text{Frob}_p)^d = \text{id}$. However, if $(\text{Frob}_p)^m = \text{id}$, where $m < d$, then every element of K is a root of $x^{p^m} - x$, but this polynomial has at most p^m elements in K , which is less than $p^d = |K|$. $\square$

Proposition 6.1.10

Let $K \subset L$ be an extension of finite fields of degree d . Then, L is Galois over K , with Galois group $\text{Gal}(L/K) \cong C_d$. It is generated by Frob_q , where $q = |K|$. Moreover, $K = L^{\text{Frob}_q}$ is the unique subfield of L of size q .

Proof. We know that Frob_q acts trivially on K and is an automorphism of L , so $\text{Frob}_q \in \text{Aut}_K(L)$. Recall (proposition 6.1.9) that Frob_p has order $[L : \mathbb{F}_p] = [L : K][K : \mathbb{F}_p] = d[K : \mathbb{F}_p]$. Hence, $\text{Frob}_q = (\text{Frob}_p)^d$ has order d and so generates C_d . We have $C_d \subset \text{Aut}_K(L)$. We know that $|\text{Aut}_K(L)| \leq [L : K] = d$, hence $C_d \cong \text{Aut}_K(L)$. This also proves that L/K is Galois, so $\text{Gal}(L/K) \cong C_d$. By the Galois correspondence, $K = L^{\text{Gal}(L/K)} = L^{\text{Frob}_q}$, as Frob_q generates $\text{Gal}(L/K)$. If $K' \subset L$ is a field of size q , then every element of K' is fixed by Frob_q . Thus, $K' \subset L^{\text{Frob}_q} = K$. This implies $K' = K$, since $|K| = |K'|$. $\square$

Corollary 6.1.11

Let K be a finite field with q elements (q prime). Let $f(x) \in K[x]$ be an irreducible polynomial. Let L/K be a finite extension such that $f(x)$ completely splits in L (not necessarily a splitting field). Then, Frob_q is an automorphism of L over K ; it induces a cyclic permutation of the roots of $f(x)$ in L . ($\text{Aut}_K(L)$ is generated by Frob_q)

Proof. $\text{Gal}(L/K)$ is generated by Frob_q . $f(x)$ is irreducible, so Frob_q acts on the roots transitively. Consequently, Frob_q acts on the roots as a n -cycle. $\square$

In particular, if L is the splitting field of a separable polynomial $f(x) = f_1(x) \dots f_m(x) \in \mathbb{F}_p[x]$, with irreducible factors $f_i(x)$ with degrees $\deg(f_i) = n_i$, then Frob_p is a permutation of cyclic shape $(n_1)(n_2) \dots (n_m)$.

Theorem 6.1.12

Let $f(x) \in \mathbb{Z}[x]$ be a degree n separable polynomial. Let p be a prime and $f_p(x) \in \mathbb{F}_p[x]$ be the *reduction* of $f(x)$ modulo p . Assume $\deg(f_p(x)) = n$ and $f_p(x)$ is separable. Let $\mathbb{Q} \subset K$ (respectively, $\mathbb{F}_p \subset F$) be a splitting field of $f(x)$ (respectively of $f_p(x)$). Let $Z \subset K$ (respectively $Z_p \subset F_p$) be the set of roots of $f(x)$ (respectively $f_p(x)$) in L (respectively L). Then, there is a bijection $\psi: Z \xrightarrow{\sim} Z_p$ such that under this identification, $\text{Gal}(f_p)$ is a subgroup of $\text{Gal}(f)$. More formally, there is a cyclic subgroup $C \subset \text{Gal}(f)$ and an isomorphism of groups between

$$\iota: C \xrightarrow{\sim} \text{Gal}(f_p) = \text{Aut}_{\mathbb{F}_p}(F)$$

such that for any $g \in C$, $\iota(g) \circ \psi = \psi \circ g$.

Proof. This will come from theorem 6.1.40. $\square$

In other words, we can identify Z and Z_p , and this identification makes $\text{Gal}(f_p)$ a subgroup of $\text{Gal}(f)$. Put together with corollary 6.1.11 and the discussion after it, we obtain the following:

Corollary 6.1.13

Let $f(x)$ be a separable polynomial of degree n . If $f_p(x)$ is separable of degree n and is a product of irreducible factors of degree $n_1, n_2, \dots, n_m$, then the image of $\text{Gal}(f)$ in S_n and contains a permutation of cyclic shape $(n_1)(n_2) \dots (n_m)$ (Here, $n_1 + n_2 + \dots + n_m = n$).

Remark 6.1.14 — This gives a very efficient method for calculating $\text{Gal}(f)$. Only finitely many primes don’t satisfy the assumptions. Write $f(x) = a_n x^n + \dots + a_0$. If $p \nmid a_n$, then $\deg(f_p) = n = \deg(f)$. Recall that a polynomial is separable if and only if the discriminant is nonzero. If Δ_f is the discriminant of $f(x)$, then $p \nmid \Delta_f$ implies that $f_p(x)$ is separable. This is because $\Delta_f \neq 0$ if and only if $f(x)$ is separable. Thus corollary 6.1.13 can be applied with all primes except finitely many.

Some Group Theoretical Lemmas

We’re intrigued in the structure of S_n . Recall that:

- Every permutation can be written as a product of independent cycles.
- A cycle is a product of transpositions, i.e. 2-cycles:

$$(1234\dots n) = (1n)\dots(14)(13)(12)$$

Thus, transpositions generate S_n . In fact, already $(12)(23), \dots, (n-1, n)$ generate S_n . This can be proven with induction, with the realization that $S_1 \subset S_2 \subset S_3 \subset \dots$

Lemma 6.1.15

(12) and $(123\dots n)$ generate S_n .

Proof. Denote $(123\dots n)$ by σ , and consider $\sigma^i(12)\sigma^{-i}$. We get

$$\sigma(12)\sigma^{-1} = (23), \sigma^2(12)\sigma^{-2} = (34), \dots,$$

and so these generate S_n . □

Remark 6.1.16 — In the lemma above, we can’t take arbitrary transpositions and n -cycles. We need 1 and 2 to be in the specified position of the n -cycle.

Example 6.1.17

(13) and (1234) generate $D_8 \subsetneq S_4$.

The desired situation can be attained when p is prime.

Lemma 6.1.18

Let p be a prime. Then, any transposition and a p -cycle generate S_p .

Proof. WLOG, we have (12) , and let our p -cycle be $\sigma = (173652)$ (this choice for σ is completely arbitrary). Then, some power σ^i sends 1 to 2. Since $\sigma^i \neq e$, σ^i must be a p -cycle. Now, we have (12) and $(12\dots)$, and these generate S_p by the previous lemma. □

Example 6.1.19

Let $f(x) = x^5 - x - 1$ over $\mathbb{Q}$. Then, what is $\text{Gal}(f)$? We consider the reduction modulo 2, $f_2(x)$. There are no roots in $\mathbb{F}_2$. We have $f_2(x)' = x^4 - 1$. If x_0 is a common root of $f_2(x)'$, we have

$$x_0^5 - x_0 - 1 = x_0 - x_0 - 1 \neq 0.$$

This is a contradiction, so $f(x)$ is separable over $\mathbb{F}_2$. We now ask whether f_2 has an irreducible factor of degree 2. In particular, we have

$$x^5 - x - 1 = (x^2 + x + 1)(x^3 + x^2 + 1).$$

It follows that a permutation σ of cycle shape $(2)(3)$ is in $\text{Gal}(f)$. Then, σ^3 is a transposition.

Now we move to reduction modulo 3. $f_3(x)$ has no roots over $\mathbb{F}_3$. We claim $f_3(x)$ is irreducible. Then, after some calculation with irreducible degree 2 polynomials, we can see that this is true. Thus, we get a 5-cycle.

Hence, $\text{Gal}(f) = S_5$ by lemma 6.1.18.

Proposition 6.1.20

Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of prime degree p . If $f(x)$ has exactly $p - 2$ real roots (or exactly two roots are non-real), then $\text{Gal}(f) \cong S_p$.

Proof. Let L be a splitting field of $f(x)$. Let $\alpha \in L$ be a root of $f(x)$. Since $f(x)$ is irreducible, we have $\mathbb{Q}(\alpha) \cong \mathbb{Q}[x]/(f(x))$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = p = \deg(f(x))$. The tower law implies $p = [\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides $[L : \mathbb{Q}] = |\text{Aut}_{\mathbb{Q}}(L)| = |\text{Gal}(f)|$. Cauchy's theorem says that a finite group G such that $p \mid |G|$ must contain an element of order p . But every element of order p in S_p is a p -cycle.

It remains to find a transposition. We have $\mathbb{Q} \subset L \subset \mathbb{C}$. $L/\mathbb{Q}$ is a normal extension, so if $c \in \text{Aut}_{\mathbb{R}}(\mathbb{C})$ is the complex conjugation, then, since $L/\mathbb{Q}$ is normal, $\sigma(L) = L$. Thus, σ acts on L by automorphisms. We get a homomorphism $C_2 \cong \text{Gal}(\mathbb{C}/\mathbb{R}) \rightarrow \text{Aut}_{\mathbb{Q}}(L) = \text{Gal}(f)$. Note that c switches two fixes all other roots. Thus, the image of c is a transposition. We then apply lemma 6.1.18. $\square$

Example 6.1.21

Consider $x(x - 1) \dots (x - (p - 3))(x^2 + 1)$. This is not irreducible. However, we can perturb this polynomial slightly $x(x - 1) \dots (x - (p - 3))(x^2 + 1) + \frac{1}{n}$ and use analytical arguments to see that this is irreducible, but still has these roots. If n is large enough and is irreducible, then the proposition applies.

Lemma 6.1.22

Let $G \subset S_n$ is a transitive subgroup containing a transposition and an $(n - 1)$ -cycle. Then, $G = S_n$.

Proof. WLOG, $\tau = (12)$. If σ is a $(n - 1)$ -cycle involving $2, 3, \dots, n$, then WLOG, $\sigma = (23 \dots n)$. Then, $\tau\sigma = (123 \dots n)$. However, (12) and $(12 \dots)$ generate S_n . Next, if $\sigma = (23 \dots n)$ and $\tau = (2m)$, where $m \neq 1$. Since G is transitive, there exists a $g \in G$ such that $g(2) = 1$. Then, $g\tau g^{-1} = (1m') \in G$. Consider $g\tau g^{-1}$ and σ , so we conclude as above. $\square$

Remark 6.1.23 — Lemma 6.1.22 is useful because we have proposition 5.1.5. To recall the statement, this proposition says that ‘the orbits of the action of $\text{Gal}(f)$ on the roots bijectively correspond to the roots of irreducible factors of $f(x)$. In particular, if $f(x)$ is irreducible, then the action is transitive, i.e. there is only one orbit.’

Example 6.1.24

Let $f(x) = x^6 - 12x^4 + 15x^3 - 6x^2 + 15x + 12$. Eisenstein with $p = 3$ tells us that $f(x)$ is irreducible. Next,

$$f_2(x) = x^6 + x^3 + x = x(x^5 + x^2 + 1).$$

$x^5 + x^2 + 1$ isn’t divisible by $x^2 + x + 1$ and has no root in $\mathbb{F}_2$. Thus, we have a 5-cycle. Consider now reduction with $p = 5$. Only even powers survive, and substitution $t = x^2$ gives

$$t^3 - 2t^2 - t + 2 = (t - 1)(t + 1)(t - 2).$$

We substitute back in x^2 , so $x^2 - 1$ gives two roots and $x^2 + 1$ gives two roots, but $x^2 - 2$ doesn’t give any roots. Thus, $x^2 - 2$ is irreducible. Therefore, $\text{Gal}(f)$ contains a transposition and so we conclude, by lemma 6.1.22, that $\text{Gal}(f) \cong S_6$,

§6.1.ii Dedekind’s Independence Theorem

Definition 6.1.25. A **semigroup** S is a set with a binary operation, which is associative. A homomorphism of semigroups $S_1 \rightarrow S_2$ is map preserving this binary operation.

Definition 6.1.26. If a semigroup S has a unit element, then S is called a **monoid**. A homomorphism of monoids is a homomorphisms of semigroups sending 1 to 1.

Example 6.1.27

Every group is a monoid.

Example 6.1.28

Every ring is a monoid under multiplication, e.g. $\mathbb{Z}_{\geq 0}$ is a monoid under $+$ and $\mathbb{Z}_{> 0}$ is a monoid under $\times$.

Example 6.1.29

Every field is a monoid under $\times$.

Definition 6.1.30. Let S be a semigroup and let K be a field. Then, semigroup homomorphisms from $S \rightarrow K$ are called **characters** (of S with values in K).

Example 6.1.31

If S is a group, then a character $S \rightarrow K$ land in $K^\times = GL_1(K)$. Such characters are also called *1-dimensional representations*.

Example 6.1.32

A field automorphism $K \rightarrow K$ is a character of K valued in K . This can also be interpreted as a character $K^\times \rightarrow K$ with source $K^\times$.

Example 6.1.33

The zero function $S \xrightarrow{0} K$ is a character of a semigroup S (called the **zero character**). The function sending every element to 1 is also a character (called the **trivial character**).

Remark 6.1.34 — Note that for a monoid, a character χ cannot be zero, since $\chi(1) = 1$.

Theorem 6.1.35 (Linear Independence of Characters (Dedekind's Theorem))

Let S be a semigroup and let K be a field. Let $\chi_1, \dots, \chi_n: S \rightarrow K$ be pairwise different and non-zero characters. Then, $\chi_1, \dots, \chi_n$ are linearly independent in the K -vector space of functions $S \rightarrow K$.

Proof. We induct on n . The case $n = 1$ is clear. Assume that the statement is proved for $n \leq N$. If the statement is false for $N + 1$, then there are $\lambda_i \in K$, $i = 1, \dots, N + 1$, not all of them zero such that

$$\sum_{i=1}^{N+1} \lambda_i \chi_i = 0.$$

All λ_i are non-zero, otherwise we get a contradiction with the induction assumption. If we have

$$\sum_{i=1}^{N+1} \mu_i \chi_i = 0, \quad 0 \neq \mu_i \in K,$$

we rescale to ensure $\mu_1 = \lambda_1$. Then,

$$\sum_{i=1}^N (\lambda_i - \mu_i) \chi_i = 0.$$

Unless $\lambda_i = \mu_i$, for all $i = 1, \dots, N+1$, we get a contradiction. Thus, $\lambda_1, \dots, \lambda_{N+1}$ give the unique relation, up to a common multiple. Since $\chi_1 \neq \chi_2$, then we can find $c \in S$ such that $\chi_1(c) \neq \chi_2(c)$. For every $i = 1, \dots, N+1$, we have $\chi_i(cx) = \chi_i(c)\chi_i(x)$, where x is any element of S .

$$\begin{aligned} & \sum_{i=1}^{N+1} \lambda_i \chi_i(x) = 0, \quad \forall x \in S \\ \implies & \sum_{i=1}^{N+1} \lambda_i \chi_i(cx) = 0, \quad \forall x \in S \\ \implies & \sum_{i=1}^{N+1} \lambda_i \chi_i(c) \chi_i(x) = 0, \quad \forall x \in S \end{aligned}$$

$(\lambda_1 \cdot \chi_1(c), \lambda_2 \cdot \chi_2(c), \dots, \lambda_{N+1} \cdot \chi_{N+1}(c))$ is not a multiple of $(\lambda_1, \lambda_2, \dots, \lambda_{N+1})$ since $\chi_1(c) \neq \chi_2(c)$. this is a contradiction. $\square$

§6.1.iii Frobenius Lifting Theorem

Definition 6.1.36. An **algebraic integer** is a root of a monic polynomial with integer coefficients.

Remark 6.1.37 — For comparison, an algebraic number is a root of a polynomial with rational coefficients.

Every element of a finite extension of $\mathbb{Q}$ is an algebraic number, but not necessarily an algebraic integer.

Proposition 6.1.38

An element of a finite extension of $\mathbb{Q}$ is an algebraic integer if and only if its minimal polynomial has integer coefficients.

Proof. Suppose $f(x) \in \mathbb{Z}[x]$ and is monic, such that $f(\alpha) = 0$. Let $g(x)$ be the minimal polynomial of α . Therefore, $g(x)$ divides $f(x)$. We have $g(x) \in \mathbb{Q}[x]$. By Gauss's lemma, this implies that $g(x) \in \mathbb{Z}[x]$. $\square$

Proposition 6.1.39

If α and β are algebraic integers, then $\alpha + \beta$ and $\alpha\beta$ are also algebraic integers.

Proof. This won't be proven here. $\square$

Theorem 6.1.40 (Frobenius Lifting Theorem)

Let $f(x)$ be a degree n , monic, separable polynomial with coefficients in $\mathbb{Z}$, with a splitting field K . Let $Z = \{\lambda_1, \dots, \lambda_n\} \subset K$ be the set of roots of $f(x)$. Let p be a prime number, let $f_p(x)$ be the reduction of $f(x) \bmod p$. Assume that $\deg(f_p(x)) = \deg(f(x)) = n$ and $f_p(x)$ is separable. Let F be a splitting field of $f_p(x)$, and let $Z_p \subset F$ be the roots of $f_p(x)$. Define $R = \mathbb{Z}[\lambda_1, \dots, \lambda_n] \subset K$.

1. There is a ring homomorphism $\psi: R \rightarrow F$ giving a bijection $Z \xrightarrow{\sim} Z_p$:

$$\begin{array}{ccc} R & \longrightarrow & F \\ \cup & & \cup \\ Z & \xrightarrow{\sim} & Z_p \end{array}$$

2. A function $\phi: R \rightarrow F$ is a ring homomorphism if and only if $\phi = \psi \circ \sigma$, where $\sigma \in \text{Aut}_{\mathbb{Q}}(K) = \text{Gal}(K/\mathbb{Q})$.
3. There is a $\sigma \in \text{Aut}_{\mathbb{Q}}(K) = \text{Gal}(K/\mathbb{Q})$ such that $\text{Frob}_p \circ \psi = \psi \circ \sigma$. This σ is called a **Frobenius lift**.

Proof.

Claim 6.1.41. As an abelian group for $+$, $R \cong \mathbb{Z}^r$, for some $r \geq 1$, a finitely generated free abelian group.

Proof of Claim 6.1.41. $f(\lambda_i) = 0$ implies that λ_i^n is a $\mathbb{Z}$ -linear combination of $1, \lambda_i, \dots, \lambda_i^{n-1}$. Thus, every monomial $\lambda_1^{a_1} \dots \lambda_n^{a_n}$ is a $\mathbb{Z}$ -linear combination of monomials $a_i \leq n-1$. Hence, the abelian group R is finitely generated. Next, $R \subset K \cong \mathbb{Q}^{[K:\mathbb{Q}]}$, so R is torsion-free. Then, by the classification of finitely generated abelian groups, $R \cong \mathbb{Z}^r$, for some $r \geq 1$. $\square$

Claim 6.1.42. A $\mathbb{Z}$ -basis $u_1, \dots, u_d$ of R is also a basis of the $\mathbb{Q}$ -vector space K .

Proof of Claim 6.1.42. $\text{Span}(u_1, \dots, u_d)$ contains all monomials in λ_i s; hence, we have $\text{Span}(u_1, \dots, u_d) = K$. Further, $u_1, \dots, u_d$ are linearly independent: clear denominators of a $\mathbb{Q}$ -linear dependence to get a $\mathbb{Z}$ -linear dependence. Next, $\mathbb{Q}[\lambda_1, \dots, \lambda_n] \subset K$ is a subring; thus a field. (In general, for $K \subset L$ a finite extension of fields, and $b \in L$, the ring $K[b]$ is already a field since it is isomorphic to $K[x]/(g(x))$ for g the minimal polynomial of b) Since K is generated by the λ_i , we get $\mathbb{Q}[\lambda_1, \dots, \lambda_n] = K$, proving the claim. $\square$

We're now ready to prove statements 1 and 2.

1. Proposition 6.1.39 implies that every element of R is an algebraic integer. However, $\frac{1}{p} \in \mathbb{Q}$ is not an algebraic integer. Indeed, the minimal polynomial is $x - \frac{1}{p} \notin \mathbb{Z}[x]$. Thus, $\frac{1}{p} \notin R$. Thus, $p \in R$ is not an invertible element of R . Recall $pR = R \iff p \in R^\times$. Since $p \notin R^\times$, we know that $pR \neq R$. $R \cong \mathbb{Z}^d$, where $d = [K:\mathbb{Q}]$. Thus, $R/pR \cong (\mathbb{Z}/p)^r$ is finite. Thus, R/pR is a finite ring. Take any maximal ideal in R/pR . Its inverse image in R is a maximal ideal $\mathfrak{m} \subset R$ containing pR . Define $F = R/\mathfrak{m}$ (we'll show that F is a splitting field of $f_p(x)$). Let $\psi: R \rightarrow R/\mathfrak{m} = F$. $\mathfrak{m} \subset R$ is maximal, so F is a field. $\mathbb{Z} \subset R$, so $\psi(\mathbb{Z}) \subset \psi(R) = F$. Since $p \in \mathfrak{m} = \ker(\psi)$, we see that $\psi(\mathbb{Z}) = \mathbb{F}_p = \mathbb{Z}_p$. Therefore, F is a finite extension of $\mathbb{F}_p$. We extend ψ to a homomorphism $R[x] \rightarrow F[x]$,

x goes to x . Then, $f(x) = \prod_{i=1}^n (x - \lambda_i)$ is sent by ψ to $f_p(x) = \prod_{i=1}^n (x - \psi(\lambda_i))$. Since $f_p(x)$ is separable, the roots $\psi(\lambda_i)$ are distinct, for $i = 1, \dots, n$. Thus, ψ induces a bijection $Z \rightarrow Z_p$. R , as a ring, is generated by $\lambda_1, \dots, \lambda_n$. Hence, $\psi(R) = F$ as a ring, is generated by $\psi(\lambda_1), \dots, \psi(\lambda_n)$ over $\mathbb{F}_p$. Since $f_p(x)$ completely splits in F , we conclude that F is a splitting field of $f_p(x)$, so the first statement is proved.

2. $K/\mathbb{Q}$ is a Galois extension. Let $G = \text{Gal}(K/\mathbb{Q}) = \text{Aut}_{\mathbb{Q}}(K)$. $\sigma \in \text{Gal}(K/\mathbb{Q})$ acts on R . Then, $\psi \circ \sigma: R \xrightarrow{\sigma} R \xrightarrow{\psi} F$ is a ring homomorphism. We have $G = \{e = \sigma_1, \dots, \sigma_d\}$, where $d = [K:\mathbb{Q}]$. Define $\psi_i = \psi \circ \sigma_i$ ($\psi_1 = \psi$). Claim 6.1.42 says that $\psi_1, \dots, \psi_d$ are all ring homomorphisms $R \rightarrow F$. To prove this, suppose ψ_{d+1} is a ring homomorphism $R \rightarrow F$, not equal to any ψ_i , $i = 1, \dots, d$. Recall that $u_1, \dots, u_d$ is a $\mathbb{Z}$ -basis of $\mathbb{R}$. Consider the system of linear equations in $x_1, \dots, x_{d+1}$ (variable with values in F):

$$\sum_{i=1}^d \psi_i(u_j) x_i = 0.$$

Then, the j -th equation says that $\sum_{i=1}^{d+1} x_i \psi_i$ is zero at u_j . By linear algebra, there exists $x_1, \dots, x_{d+1} \in F$, not all of them 0, satisfying these equations. Then, $\sum_{i=1}^{d+1} x_i \psi_i$ is zero at each $u_1, \dots, u_d$, hence $\sum_{i=1}^{d+1} x_i \psi_i$ is zero on R , hence also on K . This contradicts Dedekind’s theorem (6.1.35). Thus, ψ_{d+1} doesn’t exist.

3. To prove this, we consider the ψ_i s as characters $R \rightarrow F$, where R is a ring, hence a monoid for multiplication. $\square$

Remark 6.1.43 — The last statement of the theorem says that under the homomorphism ψ , the Frobenius automorphism Frob_p lifts to an automorphism $\sigma \in \text{Aut}_{\mathbb{Q}}(K)$, in the sense of having a commutative diagram:

$$\begin{array}{ccc} K & \xrightarrow{\sigma} & K \\ \uparrow & & \uparrow \\ R & \xrightarrow{\sigma|_R} & R \\ \downarrow \psi & & \downarrow \psi \\ F & \xrightarrow{\text{Frob}_p} & F \end{array}$$

Remark 6.1.44 — Note that as Frob_p generates $\text{Aut}_{\mathbb{F}_p}(F)$, this immediately implies theorem 6.1.12.

§6.2 Complements on Field Extensions

§6.2.i Primitive Element Theorem

We now ask the question whether we can have one α such that

$$\mathbb{Q}(a_1, a_2, \dots, a_n) = \mathbb{Q}(\alpha).$$

That is, is there an α that generates the entire extension? The answer is, in fact, yes. That is, every finite extension of $\mathbb{Q}$ is generated by one element.

Theorem 6.2.1 (Primitive Element Theorem)

Let F/K be a finite extension of fields. Suppose $F = K(\alpha_0, \alpha_1, \dots, \alpha_n)$ such that the minimal polynomials of $\alpha_1, \dots, \alpha_n$ over K are separable. Then, $F = K(\gamma)$, for some $\gamma \in F$.

Proof. First, assume that K is a finite field. Then, $F^\times$ is cyclic. Take γ to be a generator of $F^\times$.

Now assume that K is infinite. Note that by iterating the argument, it's enough to deal with the case $n = 1$. Let $f(x)$ be the minimal polynomial of α_0 , and let $g(x)$ be the minimal polynomial of α_1 . Let L/K be a splitting field of $f(x)g(x)$. We have $K \subset F = K(\alpha_0, \alpha_1) \subset L$. Since K is infinite, we can find $c \in K$ such that $\alpha_0 + c\alpha_1 \neq \alpha'_0 + c\alpha'_1$, where $f(\alpha'_0) = g(\alpha'_1) = 0$ and $(\alpha'_0, \alpha'_1) \neq (\alpha_0, \alpha_1)$. Define $\gamma = \alpha_0 + c\alpha_1 \in F$.

Claim 6.2.2. We claim that $F = K(\gamma)$.

Proof of Claim 6.2.2. Note that it's enough to show that $\alpha_1 \in K(\gamma)$. We observe that $f(\gamma - c\alpha_1) = f(\alpha_0) = 0$. Thus, $g(x)$ and $f(\gamma - cx) \in K(\gamma)[x]$ both vanish at α_1 . In fact, α_1 is the only common root of $g(x)$ and $f(\gamma - cx)$. Since α'_1 is a root of $g(x)$, then this $\gamma = c\alpha'_1 + \alpha'_0$, if we want α'_1 to be a common root, since $\gamma - cx = \alpha'_0$. However, γ cannot equal this, so α_1 is the only common root. Thus, in L , then the greatest common divisor of $g(x)$ and $f(\gamma - cx)$ is $x - \alpha_1$. However, $g(x)$ and $f(\gamma - cx)$ are in $K(\gamma)[x]$. Calculating the greatest common divisor works the same in every field extension of $K(\delta)$. So, the greatest common divisor of $g(x)$ and $f(\gamma - cx)$ in $K(\gamma)$ must be $x - \alpha_1$. Hence, $\alpha_1 \in K(\gamma)$ so that $K(\gamma) = F$. $\square$

$\square$

Corollary 6.2.3

Every finite extension of fields of perfect fields is generated by one element.

Proof. This follows by the fact that every finite extension of perfect fields are separable. $\square$

Remark 6.2.4 — This may break down in finite characteristic.

Example 6.2.5

Consider $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. The proof suggests need to find a good c such that

$$\begin{aligned}\sqrt{2} + c\sqrt{3} &\neq -\sqrt{2} + c\sqrt{3} \\ &\neq \sqrt{2} - c\sqrt{3} \\ &\neq -\sqrt{2} - c\sqrt{3}.\end{aligned}$$

Thus, $c = 1$ works for example.

Example 6.2.6

Here’s an example of finite fields. Let $f(x) = x^3 + x + 1 \in \mathbb{F}_2[x]$, and consider the splitting field of $f(x)$, $\mathbb{F}_2[x]/(f(x))$. Note that $f(x)$ is irreducible, so this quotient is isomorphic to $\mathbb{F}_8$. We want to find a primitive element of this. In fact, $\mathbb{F}_2(\alpha)$ is valid (where α is a root of $f(x)$), as x has order 7 which is the cardinality of $\mathbb{F}_8^\times$. In fact, any element that isn’t 0, 1 works, since $\mathbb{F}_8$ is a degree 3 extension.

Exercise 6.2.7. Find a primitive element for the field extension $\mathbb{F}_{125}/\mathbb{F}_5$.

Solution: We can construct $\mathbb{F}_{125}$ as $\mathbb{F}_5[X]/(X^3 + X + 1)$. Then, a root of this polynomial, say α works (check that this has order 124), so we have $\mathbb{F}_{125} \cong \mathbb{F}_5(\alpha)$. In fact, since this is a degree 3 extension, we see that there are no subfields of $\mathbb{F}_{125}$. Thus, any element that isn’t 0, 1, 2, 3, 4 works. $\square$

§6.2.ii Normal Basis Theorem

Let F/K be a finite extension which is Galois. We ask the question: what are the Galois orbits of F ? Each orbit consists of an irreducible polynomial in $K[x]$ that splits in F . Is there an orbit consist of d elements? In fact, there exists an $\alpha \in F$ such that $\sigma_1(\alpha), \dots, \sigma_d(\alpha)$ is a basis of F .

Let L/K be a Galois extension of degree n :

$$\text{Gal}(L/K) = \{\sigma_1 = e, \sigma_2, \dots, \sigma_n\}.$$

Definition 6.2.8. A **normal basis** of L over K is a basis of L as a K -vector space which is an orbit of $\text{Gal}(L/K)$. Equivalently, we have $\alpha \in L$ such that $\alpha = \sigma_1(\alpha), \sigma_2(\alpha), \dots, \sigma_n(\alpha)$ is a basis of L over K .

Remark 6.2.9 — We can think of L as a representation of the finite group $\text{Gal}(L/K)$ over K . This says that the action of $\text{Gal}(L/K)$ on L is by K -linear transformations.

A normal basis, assuming it exists, gives L the structure of regular representation of $\text{Gal}(L/K)$: If G is finite, K is a field, then the regular representation of G over K is a vector space with basis the elements of G , so that G acts on the basis vectors by left multiplication.

Theorem 6.2.10

Every Galois extension has a normal basis.

Proof. Infinite Case: We first start with when we work over infinite fields. The primitive element theorem asserts that the existence of $\alpha \in L$ such that $L = K(\alpha)$. Let's define $\alpha_i = \sigma_i(\alpha)$. Then, $\alpha_i = \alpha$ ($\sigma_1 = e$). We claim that $\alpha_i \neq \alpha_j$ when $i \neq j$. Indeed, if $\alpha_i = \alpha_j$, then $\sigma_i(\alpha) = \sigma_j(\alpha)$ so that $\sigma_j^{-1}\sigma_i(\alpha) = \alpha$. Thus, α is fixed by a subgroup $H \subset \text{Gal}(L/K)$, such that $H \neq \{e\}$. Hence, $L = K(\alpha) \subset L^H \subsetneq L$, a contradiction. Let $f(x)$ be the minimal polynomial of α . By the claim, $\alpha_1 = \alpha_j \dots \alpha_n$ are n distinct roots of $f(x)$ in L . Hence, $f(x)$ is separable. Define $g(x) = \frac{f(x)}{(x-\alpha)f'(\alpha)} \in L[x]$. This makes sense, because $f'(\alpha) \neq 0$. Define $g_i(x) = \sigma_i(g(x)) = \frac{f(x)}{(x-\alpha_i)f'(\alpha_i)} \in L[x]$. Note that $\sigma_i(g(x))$ is the process of applying σ_i to the coefficient of $g(x)$. $g_1(x) = g(x)$. Note that $g_i(\alpha_j) = 0$, if $i \neq j$. We have $g_i(x) = \frac{\prod_{j \neq i} (x - \alpha_j)}{\prod_{j \neq i} (\alpha_i - \alpha_j)}$. Hence, $g_i(\alpha_i) = 1$. In particular, we have $g(\alpha_1) = 1$, $g(\alpha_i) = 0$ when $i \neq 1$. Let $A(x)$ be the $n \times n$ matrix with entries $A(x)_{ij} = (\sigma_i \sigma_j)(g(x))$. Define $D(x) = \det A(x)$.

We now claim that $D(x)$ is a non-zero polynomial. It's enough to prove that $D(\alpha) \neq 0$. $A(\alpha)_{ij} = g_k(\alpha)$, where $\sigma_k = \sigma_i \sigma_j$. However,

$$g_k(\alpha) = \begin{cases} 1, & \text{if } k = 1 \\ 0, & \text{if } k \neq 1 \end{cases} \iff \sigma_i = \sigma_j^{-1} \text{ in } \text{Gal}(L/K).$$

Thus, every row of $A(\alpha)_{ij}$ has a unique non-zero element. Different rows have 1s in different columns. Hence, $A(\alpha)$ is the permutation matrix encoding $g \mapsto g^{-1}$ in $\text{Gal}(L/K)$. Every permutation matrix has determinant ± 1 , hence $D(\alpha) \neq 0$.

Then, $F(x)$ has finitely many roots, so there exists $\beta \in K$ such that $D(\beta) \neq 0$. (This uses that K is infinite.) Define $\gamma_i = g_i(\beta)$. Indeed, $\gamma_i = \sigma_i(g(\beta))$, because $\sigma_i(\beta) = \beta$ as $\beta \in K$. thus, $\gamma_1, \dots, \gamma_n$ is the $\text{Gal}(L/K)$ -orbit of $g(\beta)$. It remains to prove that $\gamma_1, \dots, \gamma_n$ are linearly independent over K . For contradiction, suppose that we have $x_1, \dots, x_n \in K$ such that

$$\sum_{j=1}^n x_j \gamma_j = \sum_{j=1}^n x_j \cdot g_j(\beta) = 0.$$

Applying this to $\sigma_i \in \text{Gal}(L/K)$, we obtain

$$\sum_{j=1}^n (\sigma_i \sigma_j)(g(\beta)) \cdot x_j = 0,$$

for any $i = 1, \dots, n$. This is the same as

$$A(\beta) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0.$$

Now, $A(\beta)$ is non-degenerate because $\det A(\beta) = D(\beta) \neq 0$. Therefore, $x_1 = \dots = x_n = 0$. This shows that $\gamma_1, \dots, \gamma_n$ are linearly independent over K .

Finite Case: Next is when we work over finite fields. Let L/K be an extension of fields. Then, L/K is Galois and $\text{Gal}(L/K) \cong C_n$, where $n = [L : K]$, generated by Frob_q , where $q = |K|$. Observe that Frob_q is a linear of L as a K -vector space. Turn L into a $K[t]$ -module by setting $t \cdot y = y^q = \text{Frob}_q(y)$, where $y \in L$. Let $f(x)$ be the minimal polynomial of Frob_q . Write $f(x) = \prod_{i=1}^m f_i(x)^{n_i}$, where each $f_i(x)$ is monic and irreducible, $n_i \geq 1$. Then,

$$L \cong \bigoplus_{i=1}^m \bigoplus_{a_{ij}} K[t]/(f_i(t))^{a_{ij}}$$

as a $K[t]$ -module, where $\max(a_{ij}) = n_i$, by the primary decomposition theorem/ structure theorem for finitely generated modules over a PID.

Claim 6.2.11. For each i , there is one a_{ij} , namely n_i itself.

We will deduce this from the following claim.

Claim 6.2.12. $f(x) = x^n - 1$.

Proof of 6.2.12. The Frobenius map satisfies this:

$$(\text{Frob}_q)^n = \text{Frob}_{q^n} = \text{id}_L,$$

because $|L| = q^n$. Thus, $f(x)$ divides $x^n - 1$. To prove that $\deg(f(x)) = n$, we need to show that $(\text{Frob}_q)^0 = \text{id}_L, (\text{Frob}_q), \dots, (\text{Frob}_q)^{n-1}$ are linearly independent over K . However, $(\text{Frob}_q)^0, \dots, (\text{Frob}_q)^{n-1}$ are distinct characters $L \rightarrow L$, so they are linearly independent by the Dedekind independence theorem. Thus, $f(x) = x^n - 1$. $\square$

Proof of Claim 6.2.11. Let $\chi(x)$ be the characteristic polynomial of Frob_q . Then, $f(x) \mid \chi(x)$, so since $\deg(\chi(x)) = n$, we have $\chi(x) = f(x) = x^n - 1$. We have

$$L \cong \bigoplus_{i=1}^m \bigoplus_{a_{ij}} \underbrace{K[t]/(f_i(t))^{a_{ij}}}_{t\text{-invariant}}$$

and

$$\prod_{i=1}^m f_i(x)^{n_i} = f(x) = \chi(x) = \prod_{i=1}^m \prod_{a_{ij}} f_i(x)^{a_{ij}}.$$

Thus, we must have $a_{ij} = n_i$ (there is only one a_{ij} for each i). $\square$

By claim 6.2.11, we have

$$L \cong \bigoplus_{i=1}^m K[t]/(f_i(t))^{n_i}$$

as a $K[t]$ -module. We have ideals $(f_i(t))^{n_i} \subset K[t]$, where $f_i(t)^{n_i}$ and $f_j(t)^{n_j}$ are coprime for $i \neq j$. By the Chinese remainder theorem,

$$L \cong \bigoplus_{i=1}^m K[t]/(f_i(t))^{n_i} \cong K[t]/(f(t)) = K[t]/(t^n - 1).$$

Consider the following basis of L as a vector space over K : $1, t, \dots, t^{n-1}$. Visibly, this basis is stable under multiplication by t . This is a normal basis of L over K . $\square$

Remark 6.2.13 — The above theorem holds for both finite and infinite fields.

Remark 6.2.14 — In the proof above for finite fields, we didn't actually use the fact that it's finite. Rather, we used the cyclic part more. Thus, we can apply this proof to infinite cyclic extensions.

Example 6.2.15

Let's consider the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$. By the primitive element theorem, we may write this as $\mathbb{Q}(\sqrt{2} + \sqrt{3})/\mathbb{Q}$. However, the Galois orbit contains $\sqrt{2} - \sqrt{3}$, $-\sqrt{2} + \sqrt{3}$, and $-\sqrt{2} - \sqrt{3}$, which is of dimension 2. Thus, we try $1 + \sqrt{2} + \sqrt{3} + \sqrt{6}$. This works.

§6.3 Cyclotomic Polynomials over $\mathbb{Q}$

Definition 6.3.1. For $n > 0$ integer, the n -th **cyclotomic polynomial** is the polynomial:

$$\Phi_n(X) = \prod_{k \in (\mathbb{Z}/n\mathbb{Z})^\times} \left(X - e^{\frac{2\pi k \cdot i}{n}} \right)$$

Lemma 6.3.2

For all $n > 0$ integer,

$$\Phi_n(X) \in \mathbb{Z}[X]$$

Proof. Denote by $\mu_n \subset \mathbb{C}^\times$ the group of roots of unity and by $\mathbb{Q} \subset \mathbb{Q}(\mu_n)$ the splitting field of the polynomial $X^n - 1 \in \mathbb{Q}[X]$; a priori $\Phi_n(X) \in \mathbb{Q}(\mu_n)[X]$.

It is easy to argue that $\Phi_n(X) \in \mathbb{Q}[X]$. Indeed, the Galois group G of the extension is a subgroup of $\text{Aut} \mu_n = (\mathbb{Z}/n\mathbb{Z})^\times$ and hence the polynomial $\Phi_n(X)$ is Galois-invariant and hence in $\mathbb{Q}[X]$.

It is a little bit more subtle to show that $\Phi_n(X)$ has integer coefficients. Fix $n > 0$. It is clear from the definition that

$$X^n - 1 = \prod_{d|n} \Phi_d(X)$$

If we assume by induction that for all $d \mid n, d \neq n$, $\Phi_d(X) \in \mathbb{Z}[X]$, then it follows from the Gauss Lemma that $\Phi_n(X) \in \mathbb{Z}[X]$. $\square$

I conclude this section with the proof of the following very deep theorem of Dedekind. I am shocked and surprised by how deep this theorem is. There really seems to be no elementary proof of the irreducibility of $\Phi_n(X) \in \mathbb{Z}[X]$ for general n (if n is prime, or the power of a prime, there is an elementary proof of irreducibility using the Eisenstein criterion).

Theorem 6.3.3

Fix an integer $n > 0$, denote by μ_n the group of n -th roots of unity, by $\mathbb{Q} \subset \mathbb{Q}(\mu_n)$ the splitting field of the polynomial $X^n - 1$, and by G_n the Galois group of the extension $\mathbb{Q} \subset \mathbb{Q}(\mu_n)$. The following equivalent facts hold:

1. The cyclotomic polynomial $\Phi_n(X)$ is irreducible;
2. $[\mathbb{Q}(\mu_n) : \mathbb{Q}] = \varphi(n)$ where

$$\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times| = |\{k \mid 0 \leq k < n \text{ and } \text{hcf}(k, n) = 1\}|$$

is Euler's function;

3. The canonical injective group homomorphism $\rho : G_n \rightarrow \text{Aut} \mu_n = (\mathbb{Z}/n\mathbb{Z})^\times$ is an isomorphism.

Proof. It is easy to see that the three statements are equivalent. I prove statement (3). To show that the inclusion

$$\rho : G_n \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times$$

is surjective, we need to construct enough elements of G_n that they generate all of $(\mathbb{Z}/n\mathbb{Z})^\times$, and this we do by Frobenius lifts. The idea is clear enough and the implementation, as we shall see momentarily, straightforward.

Below for a ring A we write:

$$\mu_n(A) = \{z \in A \mid z^n = 1\}$$

Note that this assignment is functorial: if $\psi : A \rightarrow B$ is a ring homomorphism, then it induces a group homomorphism $\psi : \mu_n(A) \rightarrow \mu_n(B)$.

If p is a prime not dividing n , then — by the Jacobian criterion — $X^n - 1 \in \mathbb{F}_p[X]$ is separable, and hence Theorem 54 applies. We work with the notation of Theorem 54; in particular,

$$R = \mathbb{Z}[\zeta_n] \text{ where } \zeta_n = e^{\frac{2\pi i}{n}}$$

$\mathbb{F}_p \subset F$ is the splitting field of $X^n - 1 \in \mathbb{F}_p[X]$, and $\psi : R \rightarrow F$ the ring homomorphism whose existence is proved in Theorem 54. According to that theorem ψ gives a set-theoretic bijection from the set of roots of $X^n - 1$ in $\mathbb{Q}(\mu_n)$ to the set of roots of $X^n - 1$ in F ; these sets are the groups $\mu_n = \mu_n(R)$ (recall that R is the $\mathbb{Z}$ -subalgebra of K generated by the roots of $X^n - 1$ in K) and $\mu_n(F)$ and by what we said above about functoriality of μ_n

$$\psi : \mu_n(R) \rightarrow \mu_n(F)$$

is a group homomorphism. But we just said that it is also a set-theoretic bijection, therefore it is a group isomorphism. Because $\mu_n(R)$ is a cyclic group of order n (generated, for instance, by $\zeta_n = e^{\frac{2\pi i}{n}}$), this implies that $\mu_n(F)$ is also cyclic of order n (every finite group of the multiplicative group of a field is cyclic). For both groups, the group of automorphisms is canonically $(\mathbb{Z}/n\mathbb{Z})^\times$. The point is that the following for C_n , we have not chosen a generator. An element $k \in (\mathbb{Z}/n\mathbb{Z})^\times$ acts on C_n as $g \mapsto g^k$. This gives $(\mathbb{Z}/n\mathbb{Z})^\times = \text{Aut} C_n$.

Consider the Frobenius automorphism $\text{Frob}_p : F \rightarrow F$. Frob_p acts on $\mu_n(F)$ as the group automorphism

$$\text{Frob}_p : z \mapsto z^p$$

By Theorem 54 there exists $\sigma \in G_n$ such that $\text{Frob}_p \psi = \psi \sigma$, and σ acts on $\mu_n = \mu_n(R)$ also as $z \mapsto z^p$.

To summarise we have shown: for every prime p not dividing n , there is an element of G_n that acts on μ_n as $z \mapsto z^p$.

All we need is to show that $(\mathbb{Z}/n\mathbb{Z})^\times$ is generated as a group by the classes of primes p such that $\text{hcf}(p, n) = 1$, and this is completely obvious: take any $k \in (\mathbb{Z}/n\mathbb{Z})^\times$ and decompose it into primes: by the argument with Frobenius lifts all those primes are in $G_n \subset (\mathbb{Z}/n\mathbb{Z})^\times$, hence also $k \in G_n$, hence $G_n = (\mathbb{Z}/n\mathbb{Z})^\times$. $\square$

§6.4 Kummer Theory

Theorem 6.4.1

Let $n > 0$ be an integer, and let K be a field of characteristic $p \nmid n$ (In particular characteristic zero is okay). containing all roots of the polynomial $X^n - 1 \in K[X]$. For a field extension $K \subset L$ the following are equivalent:

1. For some $0 \neq a \in K$, $L = K(\sqrt[n]{a})$; in other words $L = K(\alpha)$ where α is a root of the polynomial $X^n - a \in K[X]$.
2. $K \subset L$ is a normal and separable extension with Galois group cyclic of order dividing n .

Proof. If $\alpha^n = a$ and $L = K(\alpha)$, then the polynomial

$$X^n - a = \prod_{\varepsilon^n=1} (X - \varepsilon\alpha)$$

is separable and L is the splitting field, hence $K \subset L$ is normal and separable. Denoting by G the Galois group, we will shortly define an injective group homomorphism $\rho : G \rightarrow \mu_n(L)$. Since $\mu_n(L)$ is a cyclic group of order n (or more generally since every finite subgroup of the multiplicative group of a field is cyclic), this shows that G is a cyclic group of order dividing n . The group homomorphism ρ is defined as follows

$$\text{for } \sigma \in G, \text{ define: } \rho(\sigma) = \frac{\sigma(\alpha)}{\alpha}$$

It is easy to see that ρ does not depend on α and that ρ is an injective group homomorphism. As I said, this concludes the proof that (i) implies (ii).

For the (deeper) converse implication, suppose that $k \mid n$ and that $K \subset L$ is a normal and separable extension with Galois group a cyclic group C_k . Let $\sigma \in C_k$ be a generator, and let $\zeta \in \mu_k(L) \subset \mu_n(L)$ also be a generator. First we will find $\alpha \in L$ nonzero such that $\sigma(\alpha) = \zeta\alpha$. By the Dedekind independence Theorem 53, for some $x \in L$ the expression

$$\alpha = \sum_{i=0}^{k-1} \zeta^{-i} \sigma^i(x) \in L$$

is nonzero, and then note that

$$\sigma(\alpha) = \sum_{i=0}^{k-1} \zeta^{-i} \sigma^{i+1}(x) = \zeta \sum_{i=1}^k \zeta^{-i} \sigma^i(x) = \zeta\alpha$$

(because the value of $\zeta^{-i}\sigma^i(x)$ only depends on $i \bmod k$). To conclude, we will show that $L = K(\alpha)$ and that $\alpha^n \in K$. The second of these assertions is obvious:

$$\sigma(\alpha^n) = \zeta^n \alpha = \alpha$$

that is, α^n is C_k -invariant and hence it lies in K . For the first assertion, the group C_k acts faithfully on the field $K(\alpha)$ (for example because the $\sigma^i(\alpha) = \zeta^i \alpha$ are all distinct as $i = 0, \dots, k-1$) hence

$$k = [L : K] = [L : K(\alpha)][K(\alpha) : K] \geq [L : K(\alpha)]|C_k| \geq [L : K(\alpha)]k$$

from which we conclude that $[L : K(\alpha)] = 1$ and hence $L = K(\alpha)$. $\square$

§6.4.i Solvable Groups and Polynomials

Definition 6.4.2. A finite group G is **solvable** if there exists a chain of normal subgroups:

$$(e) = G_0 \triangleleft \dots \triangleleft G_{k-1} \triangleleft G_k \triangleleft \dots \triangleleft G_n = G, \quad \text{such that for all } k, G_k/G_{k-1} \text{ is cyclic}$$

Theorem 6.4.3

Let $n > 0$ be an integer, and let K be a field of characteristic $p \nmid n$ containing all roots of the polynomial $X^n - 1 \in K[X]$. For a separable polynomial $f(X) \in K[X]$ the following are equivalent:

1. The polynomial $f(X)$ is solvable by radicals of order dividing n ;
2. The Galois group of the splitting field $K \subset L$ of $f(X)$ is solvable by a chain of normal subgroups as in definition above, where the G_k/G_{k-1} are cyclic of order dividing n .

Remark 6.4.4 — It follows from theorem 6.4.3 that, for all integers $n \geq 5$, there is no formula for solving all polynomials of degree n by radicals. In other words, there is no generalization to $n \geq 5$ of the quadratic, cubic, and quartic formulas that you know.

Proof Sketch. We’ll provide a sketch of the argument here, leaving details as an exercise. By a result of algebra, for all $n \geq 5$ the symmetric group S_n is not solvable.

On the other hand, by the method of Frobenius lifts, for all integers $n \geq 1$ we can easily construct degree n polynomials $f(X) \in \mathbb{Q}[X]$ such that the Galois group of the splitting field of $f(X)$ over $\mathbb{Q}$ is S_n .

Alternatively, it can be shown that for all n and for all fields K ,

$$K(X_1, \dots, X_n)^{S_n} = K(\sigma_1, \dots, \sigma_n)$$

where $\sigma_1, \dots, \sigma_n \in K(X_1, \dots, X_n)$ are the elementary symmetric polynomials. Thus the Galois group of the extension

$$K(\sigma_1, \dots, \sigma_n) \subset K(X_1, \dots, X_n)$$

is S_n , and hence the polynomial

$$f(X) = X^n - \sigma_1 X^{n-1} + \dots \pm \sigma_n = (X - X_1) \cdots (X - X_n) \in K(\sigma_1, \dots, \sigma_n)[X]$$

is not solvable by radicals. $\square$

Proof of Theorem 6.4.3. The proof is a more-or-less direct (but not super-trivial) consequence of Theorem 65: for clarity I spell it out in some detail.

Let us show that (i) implies (ii). To say that the polynomial $f(X)$ is solvable by radicals of orders dividing n is the same as to say that the roots of $f(X)$ can be computed by means of field operations involving (iterated) radicals of order dividing n . In other words there is a field extension $\Omega \supset K$, and a tower of extensions:

$$\Omega = \Omega_0 \supset \cdots \supset \Omega_{k-1} \supset \Omega_k \supset \cdots \supset \Omega_n = K,$$

where $\Omega_{k-1} = \Omega_k(\sqrt[n_k]{a_k})$ for some $n_k \mid n, a_k \in \Omega_k$ such that all the roots of $f(X)$ lie in Ω . By the assertion “(i) implies (ii)” of theorem 6.4.1, every intermediate extension $\Omega_{k-1} \supset \Omega_k$ is normal and separable with Galois group a cyclic group of order dividing n_k . The extension $\Omega \supset K$ is not necessarily normal but, by taking n_k -th roots of more elements, we can make a new $\Omega \supset K$ with the same properties that is also normal and separable. Working with this new Ω , denote by $\tilde{G}$ the Galois group of $\Omega \supset K$. Letting $\tilde{G}_k = \Omega_k^\dagger$, we have a chain of subgroups:

$$(e) = \tilde{G}_0 \triangleleft \cdots \triangleleft \tilde{G}_{k-1} \triangleleft \tilde{G}_k \triangleleft \cdots \triangleleft \tilde{G}_n = \tilde{G}$$

such that for all k , $\tilde{G}_k/\tilde{G}_{k-1}$ is cyclic of order $|n_k|$. Consider now the splitting field $L \supset K$ of our original polynomial $f(X) \in K[X]$. Since L is generated by the roots of f , $\Omega \supset L \supset K$. Because $L \supset K$ is a normal field extension, $H = L^\dagger$ is a normal subgroup of $\tilde{G}$ and the Galois group of $L \supset K$ is $G = \tilde{G}/H$. Denote by G_k the image of $\tilde{G}_k$ in G . Then

$$G_k = \tilde{G}_k/(H \cap \tilde{G}_k)$$

and the natural group homomorphism $\tilde{G}_k/\tilde{G}_{k-1} \rightarrow G_k/G_{k-1}$ is surjective. It follows that G_k/G_{k-1} is a cyclic group of order dividing n_k , and this shows that (i) implies (ii).

The reverse implication “(ii) implies (i)” follows directly from the assertion “(ii) implies (i)” of theorem 6.4.1. $\square$

7 Appendix

Proposition 7.0.1 (Gauss's Lemma)

Let $f(x) \in \mathbb{Z}[x]$. If $f(x) = g(x)h(x)$, where $g, h \in \mathbb{Q}[x]$, then $\exists a, b \in \mathbb{Q}$ such that $ag(x) \in \mathbb{Z}[x]$ and $bh(x) \in \mathbb{Z}[x]$, and $f(x) = (ag(x))(bh(x))$.

Proof Sketch. Let n be the lcm of the denominators of the coefficients of $g(x)$, and let m be that for $h(x)$. Now, define $d = nm$. Note that $df(x) = (ng(x))(mh(x))$, so all components are in $\mathbb{Z}[x]$. If $d = 1$, then we're done. Otherwise, $\exists p$ prime such that $p \mid d$. Then, it follows that either $ng(x)$ or $mh(x)$ is zero mod p , so WLOG, assume that this is $ng(x)$. Thus, $p \mid ng(x)$, and so we can divide out primes. $\square$

Recall the fact that for polynomials of degree less than or equal to 3 over a field k , then irreducibility is equivalent to having no roots in k .

Example 7.0.2

We can apply the above to $x^3 + 5x + 1$. This is irreducible over $\mathbb{Q}$ if and only if there are no roots in $\mathbb{Q}$. By Gauss's lemma, we know that if there is a root in $\mathbb{Q}$, then there are a, b such that

$$x^3 + 5x + 1 = (ax + b)q(x),$$

where $q(x) \in \mathbb{Z}$ and quadratic. Firstly, we get that $a = 1$ and the leading coefficient in $q(x)$ is 1. This root divides 1, so it is ± 1 .

Proposition 7.0.3 (Eisenstein's Criterion)

Let $f(x) = a_n x^n + \dots + a_1 x + a_0$, $a_i \in \mathbb{Z}$. Suppose p is a prime such that

- $p \mid a_0, \dots, a_{n-1}$ and $p \nmid a_n$
- $p^2 \nmid a_0$

Then, $f(x)$ is irreducible over $\mathbb{Q}$.

Proof Sketch. By Gauss's lemma, if $f(x)$ is reducible over $\mathbb{Q}$, then $f(x) = g(x)h(x)$, with $f(x), g(x) \in \mathbb{Z}[x]$. Then modulo p , we have that $f(x) \equiv a_n x^n \pmod{p}$. Then, we have

$$(b_r x^r)(c_{n-r} x^{n-r}) \equiv g(x)h(x) \equiv f(x) \equiv a_n x^n \pmod{p}.$$

Therefore, b_0, c_0 are multiples of p , so $a_0 = b_0 c_0$ is divisible by p^2 , which is a contradiction. $\square$

Example 7.0.4

Take $x^{10} + 5x^7 + 5$. Then, take $p = 5$. Thus, Eisenstein's criterion tells us that it's irreducible.

Example 7.0.5

Let p be prime, and consider $x^p - 1$. Then,

$$\frac{x^p - 1}{x - 1} = 1 + x + \cdots + x^{p-1} \in \mathbb{Z}[x].$$

We claim that this is irreducible. Let $t = x - 1$. Then,

$$\frac{x^p - 1}{x - 1} = \frac{(t + 1)^p - 1}{t} = t^{p-1} + pt^{p-2} + \cdots + p.$$

Thus, Eisenstein applies and the polynomial above is irreducible.

If $\text{char}(k) = p > 0$, then there exists polynomials with multiple roots which are reducible.

Example 7.0.6

Suppose $k = \mathbb{F}_p[t] = \left\{ \frac{f(t)}{g(t)} \mid f, g \in \mathbb{F}_p[t] \right\}$. Consider $x^p - t$. Then, its derivative

$$(x^p - t)' = 0,$$

as the characteristic is p . Consider now the field extension $k \subset k[\sqrt[p]{t}] = k[x]/(x^p - t)$. Indeed, Eisenstein implies that $x^p - t$ is irreducible. $t^2 \nmid a_0$, and $t \mid a_0, \dots, a_{n-1}$, and $t \nmid a_n$. In $k[\sqrt[p]{t}]$, t is a p -th power, $p = (\sqrt[p]{t})^p$. Then,

$$x^p - t = x^p - (\sqrt[p]{t})^p = (x - \sqrt[p]{t})^p$$

Example 7.0.7

$x^p + tx + t$ is a similar example to the previous. Thus, by Eisenstein, it is irreducible once more. The derivative of this is now t .